Carbon Footprint of Place-Based Economic Policies*

Sayahnika Basu†
James Madison University

Yao Wang‡
The Ohio State University

Zhanhan Yu§
The University of Hong Kong

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Abstract

We evaluate the unintended environmental impacts of Special Economic Zones (SEZs), a place-based policy promoting economic development in India through non-energy tax breaks, on firms' energy use and carbon emissions. Using firm-level energy data and a Difference-in-Differences design with matching, we find that firms within SEZs reduce carbon emissions by 22% relative to comparable firms outside. This decline is driven mainly by the adoption of newer, cleaner capital rather than reduced output. We also find suggestive evidence of a shift away from conventional energy sources toward cleaner alternatives.

Keywords: Place-Based Policy; Tax Exemption; Carbon Emission; Energy Consumption
JEL Classification: R11, R58, O13

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†Email: basusx@jmu.edu.

‡Email: wang.16488@osu.edu.

§Email: zyu77@hku.hk.

1 Introduction

Can economic growth be achieved while sustaining the environment? This long-standing question lies at the heart of the discussions on sustainable development. While increased production is typically associated with greater energy use and pollution, economic development can also spur innovation and investment in cleaner, more energy-efficient technologies (Aghion and Howitt 2008), suggesting that growth and environmental protection need not be mutually exclusive. This issue is particularly salient for developing countries, which accounted for approximately 95% of the increase in global emissions over the past decade and nearly 75% of global greenhouse gas emissions in 2023, largely driven by rapid industrialization and sustained economic growth (Climate Leadership Council 2024).

We shed light on this critical question by examining the unintended environmental consequences of a large-scale place-based policy originally designed to spur economic growth: India's Special Economic Zones (SEZs). Like many other place-based policies widely adopted in emerging economies to foster regional development (Das and Barua 1996; Fleisher, Li, and Zhao 2010; Rothenberg, Wang, and Chari 2025), SEZs in India were explicitly created to stimulate employment, drive economic growth, and facilitate globalization. Established in the 1960s and significantly expanded in 2005, the SEZ program offers substantial tax incentives to firms operating within designated areas, providing a financial impetus for industrial growth. Empirical studies show that, despite important heterogeneity and exceptions, SEZs have played a meaningful role in promoting economic growth in India (Hyun, Ravi, et al. 2018; Gallé et al. 2024). However, this success prompts a critical question: Have these same policies also led to increased energy consumption and carbon emissions at the firm level? This question is particularly pressing given that India has emerged as one of the world's top three carbon emitters, alongside China and the United States (Investopedia 2024). Notably, the industrial sector plays a central role in this trend, accounting for 24.5% of India's total greenhouse gas emissions in 2023 (International Energy Agency 2021).

While SEZs are designed to promote economic activity, their effects on environmental performance, especially carbon emissions, are theoretically ambiguous. We formalize this ambiguity within a conceptual framework in which SEZs affect carbon emissions through two margins: total energy use and the composition of energy consumption between clean and dirty sources. SEZs influence total energy use through two competing channels. On the one hand, an *efficiency channel* operates when tax incentives facilitate capital upgrading and the adoption of energy-saving technologies (Barrows and Ollivier 2018). On the other hand, an *output channel* arises when higher production levels increase energy demand. When the *efficiency channel* dominates the *output channel*, total energy consumption declines, potentially leading to lower carbon emissions. Beyond total energy use, emissions also depend on firms' ability to adjust their energy mix. Holding total energy consumption fixed, carbon emissions can still decline if firms substitute toward cleaner energy sources. Our framework predicts that such shifts are governed by the elasticity of substitution between energy types and the fixed costs associated with adopting cleaner technologies. Consequently, emission reductions are more likely in sectors where energy substitution is relatively easy, among larger firms that can better offset upfront investment costs, and in regions with lower barriers to accessing clean energy infrastructure. The overall effect of SEZs on carbon emissions therefore reflects the combined responses along both the total energy and energy-mix margins.

The quantification of the carbon footprint of place-based policies is traditionally hindered by two significant hurdles: the difficulty of identifying comparable counterfactuals and the scarcity of reliable, firm-level data on the specific behaviors that drive emissions. We address these challenges by leveraging the rich spatial and temporal variations in the implementation of India's Special Economic Zones, utilizing a high-quality dataset that provides granular insights into firm-level production, sales, capital, and energy-specific inputs. These data allow us to move beyond aggregate estimations to directly calculate each firm's carbon emissions and investigate the underlying mechanisms, such as energy efficiency improvements and shifts in fuel composition. To identify causal effects, we adopt a difference-in-differences design combined with matching. Following Görg and Mulyukova (2024), we use official documentation on SEZ locations and sizes to construct treatment buffers that match official records, allowing us to approximate treatment assignment for firms within SEZ boundaries. We then match each treated firm to control firms located within 10 kilometers of an SEZ but outside the treatment buffer using a K-means clustering algorithm based on pre-treatment characteristics. To mitigate potential misclassifications or "spillover" effects due to boundary imprecision, we exclude firms located within a 1-kilometer "donut hole" of the SEZ boundary. By comparing changes in carbon emissions before and after SEZ notification within these matched pairs, we estimate the carbon emission impacts of SEZs while controlling for localized economic shocks.

Our empirical analysis shows that, although India's Special Economic Zones (SEZs) were designed primarily to promote economic growth, they led to a substantial reduction in firm-level carbon emissions, approximately 22% in unweighted emissions and 12% in revenue-weighted emissions, following the policy implementation. Event study estimates confirm the absence of pre-trends, with the decline in emissions becoming statistically significant three years after SEZ notification. These reductions are robust across a range of empirical strategies, including staggered difference-in-differences and propensity score matching. Importantly, the decline in emissions is not driven by a contraction in economic activity: we find no evidence of reduced output and instead observe increased adoption of new assets and physical capital among treated firms, consistent with a setting in our model in which the *efficiency channel* dominates the *output channel*. Finally, we provide suggestive evidence of a shift in energy composition, characterized by a statistically significant decline in emissions from conventional energy sources alongside a corresponding increase in emissions from renewable sources.

In line with our conceptual framework, heterogeneity analyses confirm that this environmental "double dividend" is statistically affected by firms' capacity to substitute toward cleaner energy. We find that the reduction in emissions is most pronounced in the non-manufacturing sector, where a higher elasticity of substitution allows firms to more easily transition their electricity-heavy energy mix to cleaner alternatives. Conversely, the impact is more tempered in manufacturing, where reliance on a diverse mix of fossil fuels presents higher technological hurdles. Furthermore, the policy's effectiveness is amplified in regions with expanding renewable infrastructure and among larger, high-emitting firms that possess the capital to overcome the fixed costs of green technology adoption. Together, these results demonstrate that when the *efficiency channel* and energy substitution effects outweigh the *output channel*, and when firms can shift toward cleaner energy, place-based policies can serve as a powerful tool for sustainable development.

Our study contributes to several strands of literature. First, it adds to the extensive literature on place-based economic policies. The effectiveness of place-based policies is debated with mixed results in both developing and developed country contexts (Neumark and Simpson 2015). In some contexts, policies have been successful in enhancing economic activity and promoting regional development (Wang 2013; Busso, Gregory, and Kline 2013; Kline and Moretti 2014; Ehrlich and Seidel 2018; Lu, Sun, and Wu 2023), while in others, they have shown limited or no effects (Neumark and Kolko 2010; Gobillon, Magnac, and Selod 2012; Rothenberg, Wang, and Chari 2025). In the

Indian context, SEZs have been shown to catalyze structural transformation and shift labor market dynamics (Gallé et al. 2024), though often with heterogeneous impacts on firm productivity linked to ownership and rent-seeking (Görg and Mulyukova 2024). We extend this literature by providing the first comprehensive evidence of the carbon emission externalities associated with these policies. In contrast to the null ecological effects found for other large-scale industrial policies in India (Garg and Shenoy 2021), we show that SEZs generate meaningful environmental externalities. The closest prior work is the Chinese literature on zones and pollution: Hua, Partridge, and Sun (2023) find that development zones reduce county–level PM2.5; Zhao et al. (2022) document that province–level zones increase haze pollution while national-level zones have no significant effect; and Xu, Yang, and Xiao (2023) examine SEZs and carbon neutrality outcomes. We complement this body of work by examining firm-level carbon emissions in India instead and decomposing the response into efficiency and energy-mix channels using detailed firm-level energy data.

Second, we contribute to the literature studying firm behavior and carbon abatement. We provide empirical evidence on the technique effect, a fundamental shift toward lower carbon intensity, independent of changes in the scale of production Grossman and Krueger (1991). Our findings align with Shapiro and Walker (2018), who demonstrate that the vast majority of pollution reductions in US manufacturing stem from changes in production techniques driven by environmental regulation, rather than shifts in the types of goods produced or the effects of trade. Extending this lens to a developing-country setting, we demonstrate that place-based incentives can effectively lower barriers to adopting cleaner capital and energy inputs. This allows firms to fundamentally alter how they produce rather than simply reducing how much they produce. Our evidence also aligns with the emerging framework of Emissions-Adjusted Total Factor Productivity (EA-TFP) (De Ridder and Rachel 2025), by suggesting that SEZs allow firms to optimize environmental *inputs* more efficiently than their counterparts outside the zones. Importantly, the combination of significant emissions reductions with stable output and continued capital investment in SEZs points to genuine, additional abatement, rather than the largely inframarginal responses commonly documented in carbon offset markets (Calel et al. 2025).

Finally, we contribute to the literature on reconciling economic growth with environmental sustainability. Our findings carry important implications for the design of environmental regulation and industrial policy aimed at achieving sustainable growth (e.g., Gillingham and Stock 2018; Greenstone and Jack 2015; Gugler, Haxhimusa, and Liebensteiner 2021; Dechezleprêtre, Nachtigall, and Venmans 2023; Colmer et al. 2024). In a global context where conventional air pollution regulations are substantially less stringent than the socially optimal level implied by their marginal benefits (Shapiro and Walker 2025), and where the economic costs of pollution are especially severe in the Global South (Aguilar-Gomez and Rivera 2025), our results highlight how well-designed industrial policies can help bridge the gap toward socially optimal environmental outcomes.

The rest of this paper is organized as follows. Section 2 presents the background and history of the SEZ program and energy markets in India. Section 3 builds a conceptual framework that illustrates potential underlying mechanisms. Section 4 describes the main data sources and presents summary statistics. Section 5 introduces the empirical strategies we adopt. Section 6 discusses our main empirical results and robustness exercises. Section 7 explores the mechanisms and tests the predictions of the model. Lastly, Section 8 concludes.

2 Background

SEZs in India India has undergone a period of rapid economic transformation marked by trade liberalization, rising foreign investment, and expanding service sectors since the early 2000s.

Despite these positive trends, the country continued to struggle with substantial infrastructure deficits, regulatory complexity, and regional disparities in economic development (Panagariya 2010). In response, policymakers increasingly turned to targeted interventions to foster industrial growth and enhance economic performance. One such approach involved the use of place-based policies aimed at improving the business environment and catalyzing private investment in targeted regions.

India has a long history of utilizing place-based policies. It is one of the first countries in Asia to introduce the Export Processing Zones (EPZs) in the 1960s, a place-based policy aimed at promoting exports and enhancing economic development (Ministry of Commerce and Industry, Government of India 2025; Görg and Mulyukova 2024). Until the 2000s, these zones were few in number and were exclusively owned and managed by the central government. However, the lack of modern infrastructure, unstable fiscal policy, and administrative and bureaucratic delays prompted the need for reform for businesses to develop and thrive (World Bank 2005; Ministry of Commerce and Industry, Government of India 2025). In 2005, the Special Economic Zones Act was passed, allowing for private investment and a more flexible environment with administrative and fiscal benefits. The privatization of these economic zones drastically increased the number of SEZs (see Figure A.1).

According to the SEZ Act of 2005, the stated objectives of the policy include promoting new economic activity, expanding exports, attracting both domestic and foreign investment, generating employment, and improving infrastructure (SEZ Act 2005). These goals were to be achieved through a standardized legal framework governing the establishment and operation of SEZs. Firms operating within SEZs are granted both administrative and fiscal benefits. On the administrative side, the Act provides for a "single-window clearance" system, designed to expedite approvals by consolidating them under one authority. Fiscal incentives include full income tax exemption on export income for the first five years, a 50% exemption for the next five, and an additional five-year 50% exemption on reinvested profits. SEZ units were also exempt from sales and service taxes, and from the Minimum Alternate Tax (MAT) until 2012. In addition, duty-free imports and domestic procurement of inputs are allowed under the Act.

To establish a Special Economic Zone (SEZ), developers must first obtain government approval, followed by official notification and authorization to begin operations. To obtain approval, SEZ developers must demonstrate rightful ownership of sufficiently large parcels of land, which varies depending on the industry. For example, a minimum contiguous area of 10 square kilometers is required for multi-product zones, whereas only 0.1 square kilometers is required for sector-specific zones such as IT zones. When a developer is still in the process of acquiring land, only in-principle approval can be granted. Formal approval is granted when the following conditions are met: the state government's endorsement of the project, the developer's proof of land ownership, and the state government's provision of tax exemptions, assurance of adequate infrastructure, and clearance from state regulatory bodies. Once approved, a notification authorizing the commencement of operations is issued by the board. Only at this point, investment and construction are permissible (Alkon 2018; Görg and Mulyukova 2024).

While the SEZ Act is a national-level policy, state governments play a critical role in its local interpretation and implementation. Firstly, the acquisition of land required for SEZs could be challenging (Seshadri 2011). Secondly, the state government is often responsible for providing essential infrastructure, such as electricity (Kale 2014), which has been shown to significantly impact firm productivity in India (Allcott, Collard-Wexler, and O'Connell 2016). Beyond the administrative and fiscal incentives outlined earlier, states may also introduce supplementary regulations, suggesting the possibility of heterogeneous policy impacts across regions.

India's Electricity Sector and Energy Mix Electricity is the backbone of India's energy system and the dominant input in industrial production, making the structure of the electricity sector particularly important for understanding firm-level carbon emissions. The Indian electricity sector, historically overseen by the State Electricity Boards (SEBs), has undergone two transformative policy shifts: the economic liberalization of 1991 and the comprehensive enactment of the Electricity Act of 2003 (Vardhan et al. 2024). This legislative framework sought to unbundle the sector into three distinct segments—generation, transmission, and distribution, to foster competition and private participation (Bhattacharyya 2019). Regulatory oversight is now bifurcated between the Central Electricity Regulatory Commission (CERC) and State Electricity Regulatory Commissions (SERCs), while physical load management is coordinated via a hierarchical structure of National (NLDC), Regional (RLDC), and State (SLDC) Load Despatch Centres.

As of the 2024–25 financial year, the ownership of India's installed capacity reflects a significant pivot toward private investment, with central and state government entities accounting for 25% and 24% respectively, while independent power producers (IPPs) manage 51% of total generation (Central Electricity Authority 2025). Despite rapid capacity additions in renewables, coal remains the cornerstone of the energy mix, accounting for 73% of actual generation (International Energy Agency 2024), followed by hydro (8.8%), solar (7.9%), and wind (4.6%). This reliance on fossil fuels persists even as India ranks third globally in electricity consumption and renewable energy production, with the renewable share of total energy consumption increasing from 17% in 2014 to over 22% in 2023 (Myllyvirta, Aggarwal, and Sivalingam 2025).

The transition toward a low-carbon grid is increasingly driven by climate mitigation imperatives and state-level renewable purchase obligations (RPOs). However, a critical gap remains in the intersection of environmental and industrial policy. While India's Special Economic Zones (SEZs) are designed to spur manufacturing through generous tax holidays and infrastructure support, these incentives are traditionally tied to export performance rather than environmental benchmarks (Aggarwal 2024). Importantly, the SEZ Act of 2005 does not grant firms within SEZs preferential access to cleaner electricity or exemptions from the prevailing state-level energy mix. Although SEZ developers are entitled to reliable infrastructure provision, the electricity supplied to SEZ firms is still determined by the broader composition of the state grid and is not differentiated by SEZ status (Ministry of Commerce and Industry, Government of India 2025). While the Act permits captive power generation within SEZs, it neither subsidizes renewable generation capacity nor imposes renewable purchase obligations beyond those already faced by comparable non-SEZ firms operating in the same state. Consequently, any observed transition toward cleaner energy use among SEZ firms is unlikely to arise mechanically from differential access to a cleaner electricity grid.

India's Rising Carbon Emissions Over the past two decades, India has experienced one of the fastest rates of economic growth among major economies, driven largely by industrial expansion, infrastructure investment, and urbanization (Subramanian 2019; Ghosh and Parab 2021). However, this development has come with significant environmental costs (Ghosh 2002, 2010). India is now the world's third-largest carbon emitter, with energy-related emissions more than doubling since the early 2000s (Investopedia 2024). Much of this increase stems from a heavy reliance on coal for electricity generation and the expansion of energy-intensive manufacturing sectors such as steel, cement, and chemicals (Shearer, Fofrich, and Davis 2017).

This rise in emissions has drawn increasing domestic and international attention, particularly as India seeks to balance its development goals with commitments to global climate agreements such as the Paris Accord. The government has launched several policy initiatives to promote renewable energy and to improve energy efficiency due to existing barriers to clean energy adoption (Luthra et al. 2015; Mahadevan, Meeks, and Yamano 2023). However, the carbon intensity of industrial production remains a critical concern. Given the growing environmental concerns, it is essential to examine how industrial and regional development policies, such as SEZ, interact with environmental outcomes as they present a key opportunity for reform.

3 Conceptual Framework

Our conceptual framework informs the empirical analysis by modeling firms' energy consumption decisions under the Special Economic Zone policy and generating testable predictions about its impact on carbon emissions. The framework illustrates (i) how tax exemptions associated with SEZs affect firms' energy-use choices, (ii) the conditions under which firms adopt cleaner energy sources, and (iii) how these responses vary across firms, sectors, and regions.

The framework is characterized by four key features. First, it embeds energy consumption decisions directly into firms' optimization problem. Second, it distinguishes between dirtier (conventional) energy sources with higher carbon intensity and cleaner (renewable) energy sources with lower emissions, allowing for heterogeneous adoption costs. Third, it differentiates manufacturing and non-manufacturing firms by accounting for systematic differences in energy mixes and substitution patterns across energy types (Pindyck 1979; Papageorgiou, Saam, and Schulte 2017; Sharimakin 2019). Finally, it incorporates energy-use efficiency by allowing the productivity of energy inputs to vary independently of energy quantities.

3.1 Setup

Production We consider two sectors, indexed by $s \in \{M, N\}$, where M denotes the manufacturing sector and N denotes the non-manufacturing sector. Motivated by Atkeson and Kehoe (1999) and Hassler, Krusell, and Olovsson (2012), we assume that a representative firm in sector s produces output using the following Cobb-Douglas production function

$$Y_s = A_s {K_s}^{\psi_K} (E_s A_s^E)^{\psi_E} {L_s}^{\psi_L}. \tag{1}$$

Let Y_s denote the total output of a representative firm in sector s and A_s capture total factor productivity (TFP) in that sector. Production requires capital (K_s), energy (E_s), and labor (L_s), with output shares denoted by ψ_K, ψ_E, and ψ_L, respectively. These shares satisfy $0 < \psi_K + \psi_E + \psi_L < 1$, indicating decreasing returns to scale (Fikkert and Hasan 1998; Balakrishnan, Pushpangadan, and Babu 2002). To allow for endogenous improvements in energy efficiency, we introduce a parameter A_s^E that augments the productivity of energy inputs. Increases in A_s^E reflect firms' adoption of energy-saving practices, such as upgrading equipment from energy label E to label A, as well as broader investments in energy-efficient capital, production technologies, and process improvements that raise output per unit of energy consumed.[1] Lastly, firms are subject to a constant corporate tax rate, denoted by t. We also assume perfectly competitive product and input markets. Let $P = \{p, p^K, p^E, p^L\}$ denote the price vector, where p is the output price, and p^K, p^E, and p^L are the prices of capital, energy, and labor, respectively.

[1]SEZs are also associated with agglomeration economies, network effects, and knowledge spillovers Neumark and Simpson (2015), which may independently raise energy-use productivity. Because such forces reinforce rather than compete with the capital-upgrading channel, we subsume them into A_s^E for tractability; our reduced-form estimates capture the combined response.

Energy Consumption Firms in sector s choose between two types of energy inputs: clean energy, denoted by E_s^c, which generates lower carbon emissions, and conventional (dirty) energy, denoted by E_s^d, which produces higher emissions. Total energy use is modeled as a constant returns to scale (CES) aggregate of the two energy types. Formally, the total energy input of a representative firm in sector s is given by: $E_s = \left[\delta_c (E_s^c)^{\rho_s} + \delta_d (E_s^d)^{\rho_s}\right]^{\frac{1}{\rho_s}}$, where δ_c and δ_d are CES share parameters that reflect the relative importance or preference of clean and conventional energy in the energy aggregate, with $\delta_c + \delta_d = 1$.

The parameter ρ_s governs the elasticity of substitution between clean and dirty energy in sector s. The corresponding elasticity of substitution between clean and conventional energy, denoted by σ_s, is given by $\sigma_s = \frac{1}{1-\rho_s}$. We allow this elasticity to differ across sectors. For non-manufacturing firms that primarily use electricity as their main energy source, we assume that firms make limited distinctions between electricity generated from cleaner energy sources and that generated from conventional sources. Service (non-manufacturing) sectors therefore exhibit a high elasticity of substitution between clean and conventional energy inputs. In the limiting case as $\rho \to 1$, clean and dirty energy become perfect substitutes, and total energy input simplifies to $E_N = \delta_c E_N^c + \delta_d E_N^d$. In contrast, manufacturing firms use a wider range of energy inputs, such as oil, natural gas, and coal, for distinct production processes, making substitution across energy types more limited. To support this, we show in the Appendix Figure A.3 that electricity is the primary source of carbon emissions for non-manufacturing firms (including both the financial and non-financial sectors), while fossil fuels such as oil, natural gas, and coal constitute the major sources of carbon emissions for manufacturing firms.

Lastly, we assume the energy prices are set by a competitive market. We denote the unit price of conventional energy by p^{E^d} and the unit price of cleaner energy by p^{E^c}. We assume that marginal costs of clean energy are lower than those of conventional energy, $p^{E^c} < p^{E^d}$. Moreover, we follow Allcott and Greenstone (2012), and assume that there is a fixed cost f associated with adopting clean energy.[2]

Total carbon emissions associated with energy use E_s depend not only on the overall level of energy consumption but also on the composition of energy inputs and their emissions intensities. Firms combine clean energy, E_s^c, and conventional (dirty) energy, E_s^d, which differ in their carbon intensities. Let θ_c and θ_d denote the emissions intensity per unit of clean and conventional energy, respectively, with $\theta_c < \theta_d$. Total carbon emissions for a firm in sector s are therefore given by $CE_s = \theta_c \cdot E_s^c + \theta_d \cdot E_s^d$.

Formal derivations and proofs of the comparative statics are presented in Appendix C.

3.2 Firm Optimization

Total Energy Consumption SEZs offer a tax deduction of c%, applied proportionally to the corporate tax rate t. As a result, the firm's after-tax revenue is given by $(1-t(1-c))pY_s$. Assuming that firms operate in a nested decision-making structure where they first choose the optimal input levels of capital (K^*), labor (L^*), and energy (E^*) before determining the optimal energy mix across different energy sources, we show that the tax deduction c can affect energy consumption through two channels.[3]

The first channel operates through cost minimization holding output fixed. In particular,

$$\left.\frac{\partial E_s^*}{\partial c}\right|_{\bar{Y}_s} = \left.\frac{\partial E_s^*}{\partial A_s^E}\right|_{\bar{Y}_s} \cdot \frac{\partial A_s^E}{\partial c} \propto -\frac{1}{\psi}(A_s(A_s^E)^{\psi_E})^{-\frac{1}{\psi}-1} \cdot \frac{\partial A_s^E}{\partial c}$$

[2]For example, f may reflect costs or new investments associated with purchasing solar panels. Allcott and Greenstone (2012) find that high upfront costs (e.g., for solar panels) deter firms from adopting energy-efficient technologies, despite lower operational costs, particularly for smaller firms. Additionally, Covert, Greenstone, and Knittel (2016) note that renewable energy sources typically entail high initial capital costs but lower marginal costs compared to fossil fuels.

[3]The formal derivation is provided in Appendix C.

where $\psi = \psi^K + \psi^E + \psi^L$. The effect of the tax deduction on optimal energy use therefore depends on how c affects energy-use efficiency A_s^E. Under the assumption that tax reductions relax financial constraints and facilitate investments in energy-efficient capital, we have $\frac{\partial A_s^E}{\partial c} > 0$. It then follows directly that $\left.\frac{\partial E_s^*}{\partial c}\right|_{\bar{Y}_s} < 0$. We refer to this mechanism as the *efficiency channel*. Intuitively, tax reductions enable firms to adopt more energy-efficient technologies and energy-saving practices, thereby reducing the amount of energy required to produce a given level of output.

The second channel operates through the profit maximization problem, whereby energy demand rises with output. Output can expand both because the tax deduction increases the firm's effective output price, $(1 - t(1 - c))p$, and because higher energy-use efficiency (A_s^E) lowers unit production costs. As a result, firms optimally scale up production, which in turn raises total energy demand. We refer to this mechanism as the *output channel*.

The overall effect of tax reductions on total energy consumption depends on the relative strength of the *efficiency* and *output channels*. If the *output channel* dominates, tax reductions induce firms to expand production, leading to an increase in total energy use. In contrast, if output remains largely unchanged, particularly in the short to medium run due to liquidity constraints or capacity limitations that restrict production expansion, the *efficiency channel* dominates, leading to a reduction in total energy consumption.[4] Changes in energy consumption induced by tax reductions have direct implications for firm-level carbon emissions, which depend not only on the level of energy use but also on the composition and carbon intensity of the energy mix. We next examine firms' optimal energy-mix decisions.

[4]Another caveat is that the observed output increases following a tax reduction may be partly driven by the assumed Cobb-Douglas form of the production function. In a Cobb-Douglas specification, inputs are multiplicatively related to output, so reductions in effective costs (e.g., via lower taxes) automatically translate into higher optimal output. We should recognize that in reality, however, firms may not be able to expand output easily in the short term.

Energy Mix After determining the optimal level of total energy consumption, firms in sector s choose how to allocate energy between clean and conventional sources. Specifically, firms select quantities of energy inputs E_s^x, where $x \in \{c, d\}$ denotes clean and dirty energy, respectively, to minimize total energy costs subject to the energy aggregator. Because the elasticity of substitution between clean and dirty energy differs across manufacturing and non-manufacturing firms, optimal energy-mix choices vary by firm type.

For non-manufacturing firms, clean and dirty energy are perfect substitutes, reflecting their primary reliance on electricity. As a result, these firms adopt clean energy whenever

$$(Y_N^*)^{\frac{1}{\psi}} > f \cdot \left(\frac{\delta_c \delta_d}{p^{E^d} \delta_c - p^{E^c} \delta_d} \right) \cdot \tilde{A}_s^{\frac{1}{\psi}} \cdot \Psi, \tag{2}$$

and rely on conventional energy otherwise, where f denotes the fixed cost of adopting clean energy and Ψ indexes the price and elasticity parameters appearing in the optimal output function.[5] This condition is more likely to hold when: (i) the optimal output, Y_N^*, is large—as is the case for larger or more emissions-intensive firms with higher baseline energy needs; (ii) the fixed cost, f, of adopting clean energy is low, or when firms generate sufficiently high output or profits to cover these fixed costs; and (iii) the price gap between conventional and clean energy is wide, as captured by $(p^{E^d} \delta_c - p^{E^c} \delta_d)$.

For manufacturing firms, we show that $\frac{\partial (E_M^{c*}/E_M^*)}{\partial c} = 0$, implying that the clean-energy share is invariant to the tax deduction c, and the optimal share of clean energy depends solely on the relative prices of conventional and clean energy, p^{E^d} and p^{E^c}. Consequently, unlike non-manufacturing firms, manufacturing firms adjust their reliance on clean energy only in response to changes in

[5] $\Psi = \left(\frac{p^K}{\psi^K}\right)^{-\frac{\psi^K}{\psi}} \left(\frac{p^L}{\psi^L}\right)^{-\frac{\psi^L}{\psi}} \left(\frac{p^E}{\psi^E}\right)^{-\frac{\psi^E - \psi}{\psi}}$.See Appendix C for more details.

relative energy prices, specifically, when conventional energy becomes more expensive relative to clean energy. In this setting, tax deductions affect overall energy use but have no direct effect on the composition of energy inputs.

3.3 Predictions

To summarize, SEZs can therefore affect firm's carbon emissions by influencing both total energy use and the composition of energy inputs.

First, SEZs affect firms' total energy consumption, with the direction of the effect depending on the relative strength of the *output* and *efficiency channels*. When the *output channel* dominates, total energy use increases; when the *efficiency channel* dominates, total energy use declines.

Second, SEZs influence firms' energy mix, particularly for non-manufacturing firms. Even in cases where the *output channel* dominates the *efficiency channel*, so that total energy consumption increases $\left(\partial E^*_N / \partial c > 0\right)$, SEZs can raise the likelihood that the clean-energy adoption condition is satisfied, thereby encouraging firms to switch toward cleaner energy sources. As implied by the emissions equation $CE_s = \theta_c E^c_s + \theta_d E^d_s$, total emissions CE_s may nonetheless decline if the emissions intensity of clean energy (θ_c) is sufficiently lower than that of conventional energy (θ_d).

According to condition (2), switching to clean energy is more likely for larger firms with higher Y^*. Moreover, lower unit prices of clean energy (p^{E^c}) and lower fixed adoption costs (f) further increase the likelihood that firms adopt the all-clean energy corner solution.

Put together, our conceptual model yields the following testable predictions under the specified setting:

Prediction 1. *SEZs may reduce carbon emissions even in the absence of output contractions through two key channels:*

(a) *reducing total energy consumption when the efficiency channel dominates the output channel; and/or*

(b) *shifting the composition of energy use toward cleaner energy sources.*

Prediction 2. *Holding energy market conditions constant, non-manufacturing firms are more likely than manufacturing firms to adopt cleaner energy in response to tax incentives, owing to their greater substitutability between clean and conventional energy sources, and therefore experience larger reductions in total carbon emissions.*

Prediction 3. *Larger firms are more likely to experience larger reductions in carbon emissions in response to tax incentives, as they are better positioned to absorb fixed adoption costs and overcome entry barriers associated with adopting cleaner energy technologies.*

Prediction 4. *The impact of SEZs is likely to be stronger in environments where the cost of adopting cleaner energy is lower, either because of lower unit prices or reduced fixed adoption costs, for example, in regions with better access to clean-energy infrastructure.*

4 Data

We rely on two data sources to investigate the causal impact of SEZ policies on firms' carbon emissions. The first is a panel dataset on Indian firms and the second contains information on SEZs in India.

Data on Firms The *Prowess* Database compiled by the Centre for Monitoring the Indian Economy (CMIE) includes data on all publicly traded firms and a substantial number of private firms.[6]

[6]An alternative data source is the Annual Survey of Industries (ASI), which is collected at the plant level. We rely on Prowess for three main reasons. First, ASI covers only registered manufacturing establishments and therefore excludes the services sector. Second, ASI reports geographic information only at the district level, which is too coarse for our within-grid matching design. Third, ASI does not contain the detailed energy-use variables required for our analysis.

It is a firm-level panel dataset that contains detailed information from income statements and balance sheets of companies covering more than 70% of the economic activity in India's organized industrial sector and representing 75% of all corporate taxes collected by the government (Goldberg et al. 2010). The database predominantly features large and medium-sized Indian firms. For our study, we extract information on firm attributes such as location, age, size, entity type, and industry, as well as financial data including annual total sales, expenses, profits, assets, and measures of investment and physical capital.

Since our primary variable of interest is firms' carbon emissions, we begin by extracting data on firms' annual energy consumption from the *Prowess* database. This dataset provides detailed information on energy use by source, with each energy type reported in its corresponding unit. For instance, a firm that consumes both electricity and coal in a given year would have two separate records: one indicating the number of kilowatt-hours (kWh) of electricity and another showing the metric tonnes of coal consumed. To estimate each firm's carbon emissions, we follow a method similar to Barrows and Ollivier (2018), assigning a source-specific carbon emission factor to each type of energy consumed. The emission factors used in our calculations are presented in Table B.1. By multiplying the quantity of energy used by the corresponding emission factor, we calculate the total annual carbon emissions for each firm.[7]

Notably, since electricity, both purchased and self-generated, is an important source of energy consumed by firms in our sample, we adopt an additional procedure when calculating emissions from electricity consumption, accounting for variations in the energy composition (i.e., fossil fuel-based electricity vs. non-fossil-based electricity) used in electricity generation across states and years. Specifically, rather than assigning a single emission factor to electricity, we distinguish

[7]More details regarding the calculation process can be found in the Data Appendix D.

between self-generated and purchased electricity and further differentiate electricity by generation source. For self-generated electricity, firms report a specific fuel source, allowing us to assign source-specific carbon emission factors by combining data on fuel input requirements per kWh, heating values, and fuel-specific carbon coefficients from India's electricity sector, with renewable sources such as solar and wind assumed to have zero direct emissions. These calculations are performed by fuel type and year to capture changes in generation efficiency over time. For purchased electricity, where generation sources are pooled within the grid, we compute state-year–specific emission factors using data on the energy mix and generation efficiency of state power systems, calculating emissions as a weighted average across fossil and non-fossil sources.[8] As a result, variation in emissions from purchased electricity reflects both spatial and temporal differences in the electricity generation mix and changes in generation efficiency across states and over time. More details on this procedure can be found in Appendix D.

Lastly, we construct a measure of total energy use for each firm. Firms report energy use in different physical units across fuel types, hence we first aggregate total energy consumption by converting all sources into British Thermal Units (BTU). A BTU is a standardized measure of energy content defined as the amount of energy required to raise the temperature of one pound of water by one degree Fahrenheit, allowing direct comparison and aggregation across fuels with different physical units.[9]

Data on SEZs Our second data source is the India Ministry of Commerce and Industry, specifically the Department of Commerce's website on Special Economic Zones (SEZs), where informa-

[8]Data on electricity generation, energy mix, and generation efficiency are drawn from the India Climate and Energy Dashboard (ICED): https://iced.niti.gov.in.

[9]The energy content is computed directly from fuel quantities using conversion factors between fuel units and BTU, as described in Thompson and Taylor (2008).

tion on SEZs is published periodically (Ministry of Commerce and Industry, Government of India 2025).[10] From this website, we obtained detailed data on SEZs established under the SEZ Act of 2005, including their notification dates, developer names, locations (state and address), areas, and types. Figure A.1 illustrates the evolution of the number and total area of notified SEZs since the enactment of the SEZ Act in 2005. The notification of SEZs began in April 2006, with their number rapidly increasing to 260 by the end of 2010 and gradually reaching 308 by late 2016. A brief surge in the first quarter of 2017 further raised the total to 354 by the end of 2020. The expansion in SEZ areas followed a similar pattern, with relatively larger SEZs being established between 2006 and 2010, leading to a faster increase in total area compared to the number of notifications. By 2020, the total area of SEZs reached 38,200 hectares—approximately 64% of the size of Mumbai.

Mapping Firms to SEZs We geocode each notified SEZ based on its reported address and area. Figure A.2 illustrates the geographic distribution of the notified SEZs across India, with districts shaded darker indicating a higher concentration of notified SEZs. The figure shows that SEZs are primarily concentrated in populous and developed regions, such as Mumbai, Pune, Hyderabad, and Chennai. We geocode the firm-level data based on the firms' location information and then map it to the geocoded SEZs. Section 5 provides a detailed explanation of the mapping process. Our analysis is confined to a 10 km $\times$ 10 km grid around the SEZ centroids.

A natural concern in interpreting the SEZ effect is that the treated firms in our sample are predominantly incumbents, that is, firms already operating at the eventual SEZ location prior to notification, due to the nature of our DiD design. As a result, our estimates primarily capture the response of incumbent firms rather than that of new entrants incorporated after the SEZ was established. Among the 2,492 treated firm-year observations in our analysis sample, 2,434 (97.7%)

[10] Source: https://sezindia.gov.in/notified-list-sez.

correspond to firms incorporated before the SEZ notification year at their location, while only 58 (2.3%) were incorporated afterward. Appendix Table B.5 compares the observable characteristics of these two groups. The two subgroups are statistically similar in terms of firm size, manufacturing share, non-financial services share, and financial services share. However, post-notification entrants are significantly more likely to be privately held and less likely to be publicly listed, which is consistent with the 2005 SEZ policy's emphasis on promoting private-sector participation. In addition, these new entrants are absent from the mining-construction and diversified industry categories, which together account for only 3.5% of total firms in the sample.

Descriptive Statistics Appendix Table B.2 provides the summary statistics for our study sample, which consists of 8,256 firms observed from 2000 to 2019. These firms vary in terms of entity types, industries, ages, and sizes. Specifically, approximately 35% of the firms were established after 1991, 17% between 1986 and 1990, and 27% between 1972 and 1985. The vast majority (over 80%) of these firms operate in the manufacturing sector and are publicly listed. On average, each firm emits around 21,857 metric tons of carbon annually.

5 Empirical Strategy

To identify the causal effect of the SEZ policy on firms' carbon emissions, we implement a Difference-in-Differences (DiD) estimation with matching design. This approach compares the carbon emissions of similar firms located within and outside SEZs, both before and after SEZ notification. Similar to the strategy adopted by Görg and Mulyukova (2024), we define the treatment zone as a grid centered on the provided address, with dimensions matching the land area specified in the SEZ notification, as shown by the yellow squares in Figure A.4. To minimize bias from misclassified treatment and control firms, we specify a "leave-out" zone, a 1 km × 1 km square centered on the SEZ's centroid (represented by the gray area in Figure A.4). Firms located outside the treatment grid and within the leave-out grid are excluded from the analysis. We further restrict our analysis to firms within a 10 km × 10 km grid (outlined by the purple box in Figure A.4), ensuring that our sample is geographically comparable and minimizing the influence of distant firms with potentially different characteristics.

A potential concern is that SEZs may affect nearby non-SEZ firms, which would violate SUTVA if such firms are used as controls. Our design mitigates this concern by incorporating a leave-out buffer, motivated by prior evidence that the local effects of place-based policies are highly spatially bounded.[11] Moreover, if any spillovers outside the leave-out zone affect control firms in the same direction as treated firms, the DID estimate would be attenuated toward zero because the control group would partially experience the same post-treatment impact. Thus, to the extent that such spillovers remain, our estimates should be interpreted as conservative estimates of the effect of SEZs on emissions.

Our empirical strategy relies on the assumption that, in the absence of the SEZ policy, firms within SEZs would have evolved similarly to those outside SEZs. However, as is common in place-based policy evaluations, treatment assignment is not random. Policymakers may strategically select SEZ locations, and firms may self-select into these zones based on expected benefits. Consequently, firms within SEZs differ statistically from those outside SEZs in terms of observable characteristics. As shown in Appendix Table B.3, which compares the characteristics of treated and control firms as defined above, firms in SEZs tend to be larger in size, are less likely to operate in

[11]For example, Gallé et al. (2024), using municipality-level data from India, find that the labor market effects of SEZs are concentrated within a relatively narrow radius of 10 km around SEZs. While this evidence does not rule out all spillovers in our setting, it suggests that such effects are geographically localized.

the manufacturing sector, and are more often organized as public limited companies. To address this selection bias, we complement our DiD design with a matching process to construct a comparable control group of firms outside SEZs. Specifically, we adopt the following procedure. First, we use the K-means clustering algorithm to match firms based on key attributes, including size, age, ownership, and entity type. We set the number of clusters equal to the number of SEZs and determine the starting values using a matrix containing the mean values of the observed attributes for firms located in SEZs during the years prior to SEZ establishment. The K-means algorithm then groups firms into clusters by minimizing the distance between each firm and its cluster centroid. In this way, firms that are most similar in observable characteristics are grouped together. Next, we restrict the clustering to firms located within the analysis zone (the purple box in Figure A.4) and drop firms located outside these areas. This procedure allows us to obtain pairs of firms that are similar in observable characteristics but lie on opposite sides of the SEZ boundaries.[12] This approach ensures that the control group closely mirrors the characteristics of SEZ firms. Appendix Table B.4 confirms the effectiveness of this matching process, showing that differences in age, size, entity type, and industry type are statistically insignificant in the matched sample. Additionally, to account for the staggered timing of SEZ implementation, the control group includes only firms that were never treated (Callaway and Sant'Anna 2021).

Our estimation model incorporates a comprehensive set of fixed effects to account for time-, state-, district-, and industry-specific confounding factors, as specified in the following equation:

$$Y_{it} = \sigma Treat_i + \beta Treat_i \times Post_t + \zeta_{iJ} + \tau_d + \phi_k + \kappa_{st} + \varepsilon_{it} \quad (3)$$

[12] We do not restrict the matching to a one-to-one format. In other words, the pairing can be one-to-many, such that one SEZ firm can be paired with multiple firms located outside the SEZ, as long as they are similar in terms of the observed attributes.

where i, t, s, d, and k denote firm, year, state, district, and industry respectively. The outcome variable Y_{it} is the log of annual carbon emissions for firm i in year t. $Treat_i$ is a binary indicator equal to 1 if firm i is located within an SEZ, and 0 otherwise. $Post_t$ is a binary indicator equal to 1 if year t is after the SEZ notification year for firm i, and 0 otherwise. The coefficient on their interaction, β, is the parameter of interest and captures the effect of SEZ exposure on firm outcomes. Specifically, β measures the differential change in carbon emissions following SEZ notification between firms located inside SEZs and comparable firms outside SEZs. ζ_{iJ} denotes the treatment-control pair fixed effects for firm i and the matched firms J. τ_d and ϕ_k represent the district and industry (measured by 8-digit CMIE industry code) fixed effects, respectively. To control for time-varying confounding factors, such as state-level policies that could affect firms' carbon emissions, our model also includes state-by-year fixed effects, denoted by κ_{st}. These fixed effects also absorb time-varying changes in the composition of energy sources used for electricity generation across states and districts, which may otherwise influence the carbon emissions associated with purchased electricity. Although the *Prowess* data and SEZ data extend to 2021, we restrict our sample period to 2000–2019 to avoid attributing the effects of COVID-19 to SEZs. Standard errors are clustered at the district level to account for potential spatial and temporal correlation in the error term (Wooldridge 2003; Bertrand, Duflo, and Mullainathan 2004). We show later that our results are robust to alternative choices of clustering.

To further investigate the dynamic effects of SEZs, we estimate an event-study version of equation (3), specified as follows:

$$Y_{it} = \sigma Treat_i + \alpha_m \sum_{m=6}^{2} Before_{t-m} \times Treat_i + \delta_n \sum_{n=0}^{10} After_{t+n} \times Treat_i \tag{4}$$
$$+ \zeta_{iJ} + \tau_d + \phi_k + \kappa_{st} + \varepsilon_{it}$$

In this specification, we introduce the terms Before_{t-m} and After_{t+n}, which are event-time indicators, representing m years before and n years after SEZ notification, respectively. Year $t-1$ is omitted. The coefficient α_m captures the pre-treatment trends, allowing us to test the parallel trends assumption. The coefficient δ_n measures the dynamic effects of SEZs on firms' carbon emissions over time.

6 Results

6.1 The Effect of SEZs on Firms' Carbon Emissions

DiD Results Table 1 reports the main estimation results for equation (3). Panel A of Table 1 reports the estimated effects of SEZs on firms' total carbon emissions (measured in metric tons) and total energy use (measured in BTU). We find that SEZ firms experience a reduction in carbon emissions by approximately 22%, and a reduction in total energy use by 23% relative to comparable firms outside SEZs, which are statistically significant at the 1 percent level.[13] The close similarity between the estimates in columns (1) and (2) in Panel A of Table 1 implies that essentially the entire emissions decline operates through the efficiency channel, with energy-mix shifts contributing little to the aggregate response, which we will examine further in later section.

[13]The effect size for carbon emissions is calculated as $(e^{-0.248} - 1) \times 100$.

Event Study To examine the dynamic effects of SEZ exposure and assess the validity of the parallel trends assumption, we conduct an event study analysis based on the specification in equation (4), using carbon emissions as the outcome variable. Figure 1 plots the estimated coefficients α_m and δ_n along with their 95% confidence intervals from equation (4). We find no systematic differences in carbon emission trends between firms located inside and outside SEZs prior to SEZ notification, supporting the parallel trends assumption underlying our identification strategy. Following SEZ notification, carbon emissions decline progressively over the first five years, with the effect gradually attenuating and dissipating after approximately eight years.

Robustness We assess the robustness of our results across different samples, model specifications, and clustering levels. Appendix Table B.7 presents the results of various robustness checks, all based on the preferred model specification of Table 1, using specification (3).

First, we test the sensitivity of our findings to different clustering levels. Columns (1) to (3) of Appendix Table B.7 show results with standard errors clustered at the (1) treatment-control pair, (2) state and year, and (3) firm and district-year levels, respectively. In all cases, the standard errors are consistent with those reported in Table 1, indicating that our findings are robust to alternative clustering approaches.

Second, we examine whether our results are driven by firms established during the early years of SEZ implementation (2000–2005). Given that the SEZ policy was first issued in 2000 and fully enacted in 2005, we re-estimate the model using two sub-samples: firms incorporated before 2000 and those incorporated before 2005. Columns (4) and (5) of Appendix Table B.7 confirm that the estimates remain statistically significant and consistent with the main findings. Although the magnitude of the estimates is slightly larger than in the full sample, these differences are not statistically significant.

Third, we test the robustness of our results to different spatial definitions of the treatment and leave-out zones. Specifically, we re-estimate the model using a 5 km treatment zone and a 2 km leave-out zone. Columns (6) to (8) of Appendix Table B.7 show that the estimated effects range from -0.2 to -0.3, which are statistically indistinguishable from our main results. This consistency demonstrates that our findings are not sensitive to the spatial extrapolation of the SEZ treatment area.

Fourth, as an alternative to our matched DiD specification, we implement a staggered DiD estimator with inverse probability weighting (IPW) following Callaway and Sant'Anna (2021). Appendix Figure A.8 illustrates the results using IPW. While the precision of the estimates varies slightly, the direction of the estimated effects and the lagged effect pattern remain consistent with our main findings.

Finally, another potential concern relates to the accuracy of self-reported energy data. Because Indian firms can typically claim input tax credits for taxes paid on their purchases, including energy inputs, firms may differ in their incentives to report energy consumption accurately. For example, firms outside SEZs may have stronger incentives to report accurately in order to avoid overpaying taxes, whereas firms benefiting from SEZ tax incentives may face weaker reporting incentives. This asymmetry could generate systematic differences in reporting behavior and in the incidence of missing energy data between treated and control firms, potentially introducing sample selection bias.

We examine this concern by analyzing whether SEZ and non-SEZ firms exhibit different trends in accurately tracking their energy consumption before and after the establishment of SEZs. To proxy for potential reporting inaccuracies, we classify observations with missing values as instances of insufficient tracking. These missing values arise either from ambiguous energy source names or unit measures in the energy consumption table, or from missing consumption quantities required to compute carbon emissions. Using this definition, we construct a binary indicator equal to one if any such missing value is present in a firm-year observation. We then replicate our baseline analysis using this binary indicator as the outcome variable. The results for this exercise are reported in Appendix Figure A.9. The event-study results show that the rate of inaccurate reporting increases slightly for SEZ firms after notification. However, this effect disappears after the first period and is not statistically different from zero from three years after notification onward, the periods in which we observe effects on emissions in our main analysis. We therefore interpret these results as indicating no systematic effect of SEZs on the accuracy of energy use reporting. As an additional robustness check, we replicate the main event study analysis on a restricted sample consisting only of firms that consistently and accurately report their energy consumption throughout the sample period. The corresponding results are reported in Appendix Figure A.10. As shown, these results remain qualitatively and quantitatively similar to our main findings.

6.2 Impacts on Profits, Sales, and Expenses

To better understand the source of carbon emission reductions, we first examine whether SEZs lead to a contraction in firm production, which could mechanically reduce energy use and emissions.

In particular, we examine whether SEZs significantly affect firms' production outcomes, measured by annual total profits, total sales, and total expenses. We obtain firm level financial information from the *Prowess* database and construct a firm year panel. To maintain consistency with the main analysis, we restrict the sample to firms that report energy use, ensuring comparability between production and energy outcomes. We then re-estimate the specification in equation (3), using these financial measures as dependent variables.

Table 2 reports the corresponding estimation results, with each column indicating a different outcome variable. We find no statistically significant effects of SEZ designation on firms' production or financial outcomes. Although the point estimates for sales and profits are positive, they are not statistically significant. We also observe suggestive but imprecise evidence of a decline in fuel expenses alongside an increase in total expenses. The absence of statistically significant performance effects is consistent with prior evidence on India's SEZ program. Studying the same policy, Görg and Mulyukova (2024) document heterogeneous productivity responses across ownership types, with the weakest effects observed among publicly owned firms. Our sample is similarly dominated by publicly listed firms, which account for more than 80% of observations (Table 2). Accordingly, our null results for profits and sales should not be interpreted as evidence that SEZs fail to promote economic activity more broadly. Rather, they suggest that the effects of SEZs may operate through channels other than firm-level profits and sales among the firms and years covered by our data.

Overall, the absence of significant changes, and if anything weakly positive effects, in profits, sales, and expenses indicates that the observed reductions in energy use and emissions are not driven by declines in output or economic activity. Together with the result in Column (2) of Panel A of Table 1, showing a significant reduction in total energy use following SEZ designation, these findings align with model prediction 1(a), under which SEZs reduce energy consumption without compromising output when the *efficiency channel* dominates the *output channel*.

Next, we examine the effects of SEZ designation on carbon emissions and energy use per unit of firm sales. Panel B of Table 1 reports the corresponding estimation results. The per-sales estimates are sizable and qualitatively consistent with those reported in Panel A. Specifically, we find that firms located in SEZs experience significant reductions in carbon emissions per unit of sales and energy use per unit of sales of approximately 12% and 30%, respectively, relative to comparable firms outside SEZs.[14]

6.3 Impacts on Investment, Assets, and Capital

To further examine the *efficiency channel* underlying prediction 1(a), namely, that SEZs facilitate capital upgrading and production modernization, thereby improving energy efficiency through embodied technological change, we provide additional suggestive evidence at the aggregate level. Although we do not observe energy efficiency at the capital level directly, Appendix Figure A.12 presents patterns consistent with this mechanism. While not causal, the figure documents a steady decline in total energy use per unit of GDP in India over time, indicating that comparable levels of output are produced using progressively less energy. This trend coincides with periods of rapid capital accumulation and industrial modernization, suggesting that newer capital vintages and upgraded production facilities are increasingly energy efficient in the Indian context.

To examine whether SEZs promote such capital upgrading at the firm level, we test whether SEZ designation affects firms' investment behavior and capital accumulation. We re-estimate equation (3) using firm-level measures of short-term investment, total assets, property, plant, and equipment (PPE), and land and buildings as outcome variables.

The results are reported in Table 3. The estimates indicate that SEZ designation is associated with statistically significant increases across all measures of capital accumulation. In particular, SEZ firms exhibit higher investment and larger capital stocks, with investment rising by 2.7%, total

[14]The sample size for per-unit outcomes is smaller due to missing values in the sales variable. To rule out sample selection as a driver of the difference in magnitude between level and per-unit estimates, Appendix Table B.6 reports results for outcomes measured in levels using the same sample as in Panel B of Table 1. The estimated effects on energy use and carbon emissions are closely aligned with those reported in Panel A of Table 1.

assets by about 3.4%, and PPE by about 4.0%. We also find a 3.0% increase in land and building assets, consistent with expansion or modernization of production facilities rather than downsizing or contraction.

Taken together, these findings support a mechanism in which SEZs induce capital and technological upgrading. In light of the aggregate evidence on declining energy intensity, such capital upgrading plausibly translates into improved electricity efficiency at the firm level. This interpretation aligns closely with the *efficiency channel* described in model prediction 1(a).

6.4 Changes in Energy Consumption by Category

Next, we examine the effects on different categories of energy use to identify which sources drive the main results and whether SEZs induce changes in the composition of the energy mix. Following the IPCC 2011 classification (Edenhofer et al. 2011), we first differentiate emissions by source and group them into two broad categories: conventional and renewable energy. Conventional energy includes the direct use of coal gas and oil as well as electricity generated from these sources, together with the direct use of wood, wood residues, and agricultural products. Renewable energy includes the direct use of biomass, such as biodiesel and biogas from captured methane, as well as electricity generated from biomass, hydro, solar, and wind.

We then re-estimate our main specification using source specific energy measures as outcome variables. Table 4 shows the results, broken down by conventional and renewable energy sources. Column (1) reports the results for conventional energy. We find a statistically significant decline in emissions from conventional energy among firms located within SEZs with an estimated reduction of about 23%. Column (6) reports the results for renewable energy use. Consistent with the close to zero carbon content of most renewable sources-essentially, renewable energy sources such as solar and wind are defined as zero by construction-we find no significant effects on renewable energy related emissions. However, some renewable energy like biomass does produce non-zero carbon emission. To capture changes along the extensive margin, we construct a dummy variable indicating whether there are any emissions from renewable sources. As shown in Column (7), SEZ increases the probability of adopting renewable energy by about 0.5 percentage points.

Furthermore, we disaggregate emissions from conventional energy into fossil fuels (coal, oil, and gas), wood and agricultural products, and electricity, including both self generated and purchased power. The corresponding estimates are reported in Columns (2) through (6) of Table 4. The results show that the overall reduction in conventional energy emissions is primarily driven by declines in electricity related emissions. Specifically, emissions from self generated and purchased electricity fall by approximately 21% and 19%, respectively, for SEZ firms, while we find no statistically significant effects for the direct use of other conventional energy sources.

The decline in electricity related emissions is unlikely to reflect broader decarbonization of India's power grid. SEZ and non-SEZ firms in our sample are geographically proximate and are typically connected to the same local electricity grid, and therefore face similar changes in the generation mix over time. Instead, we find that the reduction in electricity-related carbon emissions is associated with lower energy consumption. As shown in Table 5, both purchased and self-generated electricity use decline significantly among SEZ firms in the post-treatment period.

7 Heterogeneity Analysis

We further explore the mechanisms driving our results by testing the predictions outlined in Section 3 through an analysis of heterogeneous treatment effects across industries, firm sizes, and regions. These exercises provide insights into how access to clean energy, firms' long-term decision-making capabilities, and industry characteristics influence the observed reductions in carbon emissions.

Heterogeneity by Industry We begin by examining how the impact of SEZs on firm-level carbon emissions varies across industries. Using 6-digit CMIE industry codes, we classify firms into three industry groups: Manufacturing, Services, and Mining and Construction. We then estimate our preferred model separately for each of these groups. Figure 2 presents the estimation results.[15] Our findings reveal substantial and statistically significant reductions in carbon emissions for firms in the Services sectors, including both Non-Financial and Financial Services. Specifically, SEZs are associated with a 45% reduction in carbon emissions for firms in the Services sector. In contrast, the estimated effects for the Manufacturing and Mining and Construction sectors are small and statistically insignificant. These results are consistent with our theoretical prediction 2 in Section 3, which suggests that firms with greater flexibility in energy use, such as those in the Services sector, are more responsive to SEZ incentives.

Heterogeneity by Firm Size Firms of different sizes may respond differently to SEZs, leading to varied production and energy use decisions, which ultimately affect their carbon emissions. To explore this heterogeneity, we categorize firms into two groups based on size: Top 50% and Bottom 50%.[16] Estimating our preferred model separately for these two groups, we find that larger firms (Group 1) generally experience a more substantial reduction in carbon emissions in response to SEZs. As shown in Figure 3, firms in the top 50% of the size distribution within SEZs reduce

[15]Due to the small sample size, there is no estimate for the Diversified group.

[16]*Prowess* defines firm size as the three-year average of total income and total assets. The size decile variable indicates a firm's relative position in the overall distribution of companies by size. Group 1 comprises firms in Deciles 1 through 5, and Group 2 includes firms in Deciles 6 through 10. See *Prowess*'s data dictionary for more details on the construction of the firm size variable.

their carbon emissions by approximately 27%, an effect more than seven times greater than the 5% reduction observed among smaller firms in the bottom 50%. This finding is consistent with our theoretical prediction 3, which suggests that larger firms are better positioned to adopt cleaner energy due to lower effective entry barriers for adoption.

Beyond firm size, we also explore heterogeneity by emitter type. Specifically, we rank firms according to their pre-SEZ average carbon emissions and re-estimate our baseline specification separately across emitter groups. As shown in Appendix Figure A.11, the results indicate that carbon reductions are primarily driven by high-emitting firms, with the effect estimated at around 32%. This finding is broadly consistent with Prediction 3, as these firms are more likely to overcome the fixed costs associated with adopting cleaner energy and have greater potential for emission reductions.

Heterogeneity by Access to Clean Energy There are significant regional variations in access to clean energy across India. The northern, southern, and western regions have experienced rapid growth in renewable energy markets, with cleaner energy sources accounting for a larger share of their total energy mix. In contrast, the northeastern, eastern, and central regions rely more heavily on conventional, high-emission energy sources, with a much smaller presence of renewable energy. We classify the country into two clusters: Cluster 1 includes the Northern, Southern, and Western Zones, and Cluster 2 consists of North Eastern, Eastern, and Central Zones. It is also worth noting that the regions in Cluster 1 are generally more economically developed than those in Cluster 2.[17] Appendix Figure A.13 plots trends in installed electricity capacity (in MW) by energy source and region from 2015 to 2024. The figure reveals a clear increase in clean energy capacity in Cluster 1, while installed capacity in Cluster 2 remains largely unchanged across both clean and

[17]Appendix Figure A.14 provides a map of the six zonal divisions, which we collapse into two clusters.

conventional sources.[18] Overall, our findings suggest that the impact of SEZs on firms' carbon emissions is strongly influenced by regional factors, including the composition of the energy mix, access to cleaner energy sources, and the level of economic development. Consistent with our model prediction 4, SEZs are more effective at reducing carbon emissions in regions with greater access to clean energy, where firms can more readily transition to greener energy sources.

8 Conclusion

This paper examines the impact of place-based policies that offer tax incentives to firms, focusing on how these policies influence firms' energy use and carbon emissions. Although SEZs are primarily designed to stimulate economic growth, our findings reveal a significant but unintended outcome: a substantial 22% reduction in carbon emissions among firms located within SEZs compared to similar firms outside these zones. This decline in emissions is both notable and nuanced, driven by adopting newer, cleaner capital rather than a contraction in production. Our analysis shows that improvements in energy use efficiency and access to cleaner energy sources are key determinants of this reduction. Firms located in regions with rapidly expanding renewable energy infrastructure experience more substantial declines in carbon emissions. Additionally, larger firms demonstrate a greater capacity to adopt cleaner energy technologies in response to SEZ incentives, likely due to their superior production capacity and greater flexibility in adjusting energy use.

These findings present a cautiously optimistic view of the relationship between economic development and environmental sustainability. Place-based policies like SEZs have the potential to steer industrial sectors toward greener practices, promoting the adoption of advanced technologies

[18] Installed electricity capacity data are available from 2015 onward and are sourced from ICED (https://iced.niti.gov.in/energy).

and renewable energy sources. By aligning economic incentives with environmental objectives, SEZs can serve as a model for balancing growth with sustainability.

However, our study also highlights the complexity of assessing the environmental impacts of place-based policies. The heterogeneous effects observed across regions and firm types underscore the need for targeted approaches to maximize environmental benefits. Policymakers should consider regional energy availability, firm characteristics, and sectoral differences when designing and implementing place-based policies. Future research could provide more direct evidence on the role of technology adoption and further explore other mechanisms driving firms to reduce carbon emissions, particularly regulatory environments, and firm-level characteristics. Such insights will be valuable for designing industrial and environmental policies that promote sustainable economic development.

Declaration of generative AI and AI-assisted technologies in the writing process *During the preparation of this work, the authors used ChatGPT in order to proofread and improve the language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.*

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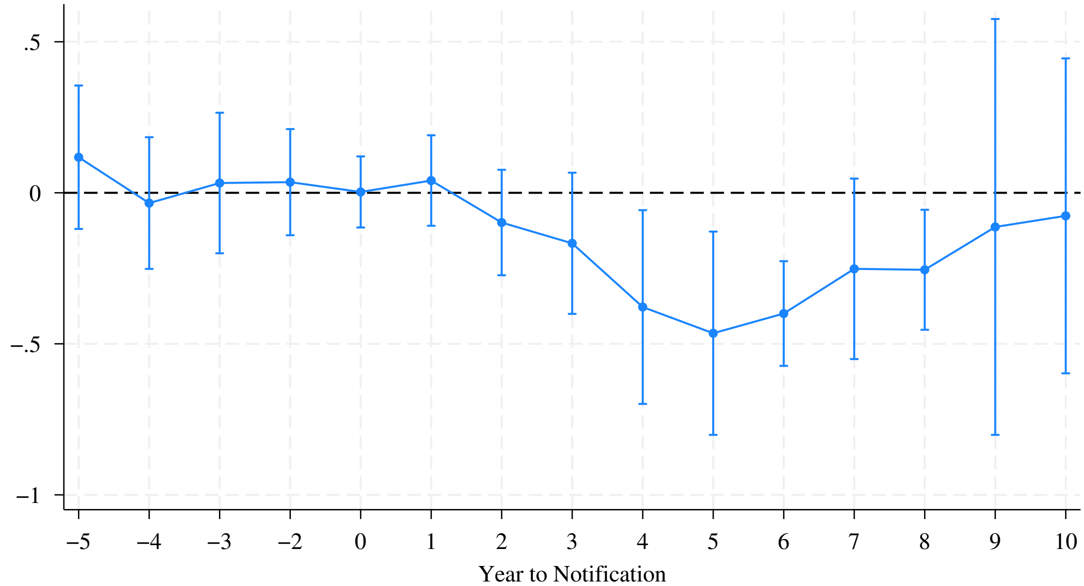


Figure 1: Event-Study Results

Notes: This figure presents the point estimates and their 95% confidence intervals for the event study estimating equation (4) where the dependent variable is log of annual carbon emissions at the firm-year level. The model includes interaction terms between SEZ treatment and time dummies indicating 1-5 years before the SEZ notification and 1-10 years after the notification, and controls for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. The standard errors are clustered at the district level.

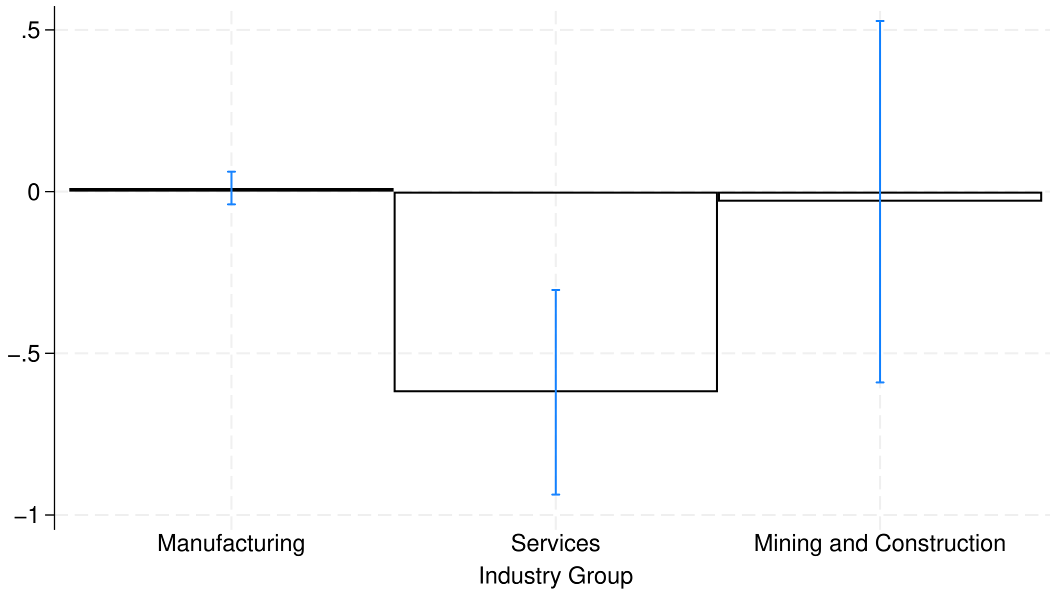


Figure 2: Heterogeneity Results: By Industry

Notes: This figure shows the point estimates for equation (3) where the dependent variable is log of annual carbon emissions and their 95% confidence intervals by industry group. We define four industry groups based on 6-digit CMIE industry codes: Manufacturing, Services (combining Financial and Non-Financial), Mining and Construction, and Diversified. The estimation is based on the preferred model controlling for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. Due to the small sample size of the Diversified group, we obtain no estimate for this group. Therefore, it is omitted from the figure. The standard errors are clustered at the district level.

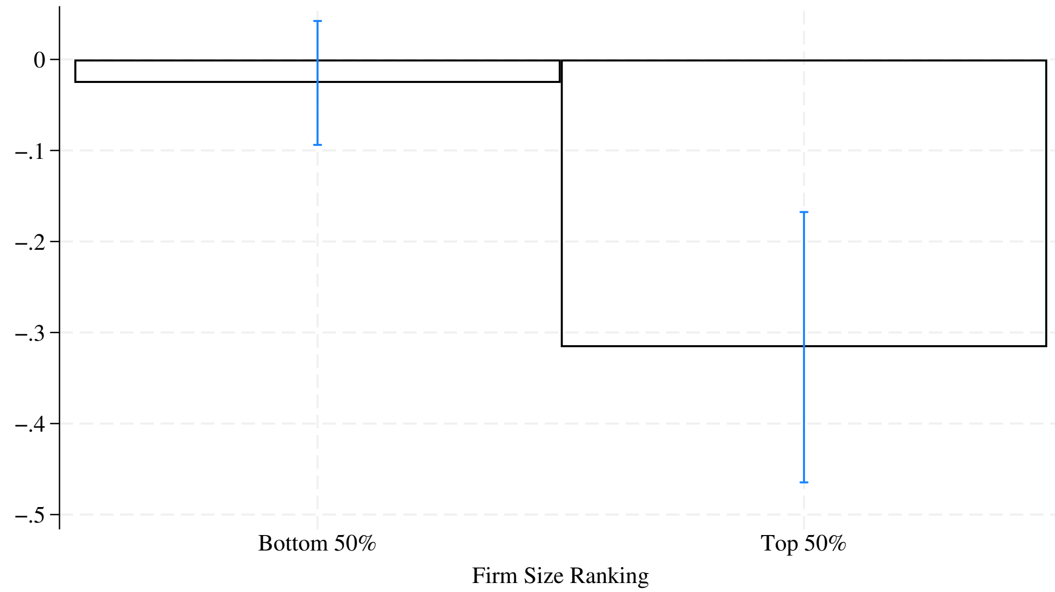


Figure 3: Heterogeneity Results: By Firm Size

Notes: This figure summarizes the point estimates for equation (3) where the dependent variable is log of annual carbon emissions and their 95% confidence intervals separately for two sub-groups based on firm size deciles: firms in Deciles 1 through 5 and firms in Deciles 6 through 10. The estimation is based on the preferred model controlling for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. The standard errors are clustered at the district level.

Table 1: Estimation Results: Carbon Emissions and Total Energy Consumption (BTU)

	(1)	(2)
Panel A:	Log(Carbon Emissions)	Log(Energy Use)
β	-0.248***	-0.255***
	(0.0904)	(0.0805)
N	33932	34144
R^2	0.64	0.63
Panel B:	Log(Carbon Emissions/Sale)	Log(Energy Use/Sale)
β	-0.127*	-0.353***
	(0.0697)	(0.1112)
N	16125	16125
R^2	0.50	0.57

Notes: This table shows the main estimation results using specification (3). * p<0.10 ** p<0.05 *** p<0.01. Standard errors clustered at the district level are reported in parentheses. All models control for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. Energy consumption is measured in millions of British Thermal Units (BTU).

Table 2: Estimation Results: Profits, Expenses, and Sales

	(1) LnTotalExpenses	(2) LnExpenses(Fuel)	(3) LnSales	(4) LnProfits
β	0.151	-0.063	0.175	0.129
	(0.119)	(0.062)	(0.132)	(0.118)
N	34294	34294	34294	32079
R^2	0.567	0.587	0.527	0.552

Notes: * p<0.10 ** p<0.05 *** p<0.01. Estimation is based on our preferred model (equation (3)). Standard errors clustered at the district level are reported in the parentheses. Expenses, profits, and sales are measured in million US$.

Table 3: Estimation Results: Investment, Assets, and Physical Capital

	(1) LnInvestment	(2) LnTotalAssets	(3) LnPPE	(4) LnLandBuilding
β	0.027*	0.033*	0.040**	0.030*
	(0.015)	(0.018)	(0.019)	(0.015)
N	34294	34294	34294	34294
R^2	0.583	0.545	0.543	0.556

Notes: * p<0.10 ** p<0.05 *** p<0.01. Estimation is based on our preferred model (equation (3)). Standard errors clustered at the district level are reported in the parentheses. Short-term Investment, total assets, PPE (property, plant, & equipment), and land and building are measured in million US$.

Table 4: Estimation Results: Carbon Emissions by Energy Source

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Conventional: Log(Emission)					Renewable	
	Overall	Fossil Fuels	Wood& Agri. Products	Electricity (self-generated)	Electricity (purchased)	Log(Emission)	Any Emission
β	-0.267***	0.031	0.026	-0.240***	-0.205*	-0.010	0.005*
	(0.092)	(0.106)	(0.020)	(0.057)	(0.106)	(0.018)	(0.003)
R^2	0.637	0.530	0.437	0.559	0.628	0.576	0.439

Notes: This table summarizes the estimation results by energy source. N = 34484. * p<0.10 ** p<0.05 *** p<0.01. Standard errors clustered at the district level are reported in parentheses. We divide emissions from conventional energy by sub-category: fossil fuels such as coal, oil, and gas, wood and agricultural products, self-generated electricity, and purchased electricity. The estimation is based on the preferred model controlling for year, state, treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects.

Table 5: Estimation Results: Electricity Consumption by Type

	(1)	(2)	(3)	(4)
	Log(Electricity Consumption in Kwh)			
	Purchased	Self-generated		
		Overall	from Conventional	from Renewable
β	-0.235**	-0.391***	-0.433***	-0.211
	(0.099)	(0.133)	(0.157)	(0.298)
N	33286	21943	21691	1018
R^2	0.631	0.667	0.667	0.783

Notes: * p<0.10 ** p<0.05 *** p<0.01. Estimation is based on our preferred model (equation (3)). Standard errors clustered at the district level are reported in the parentheses.

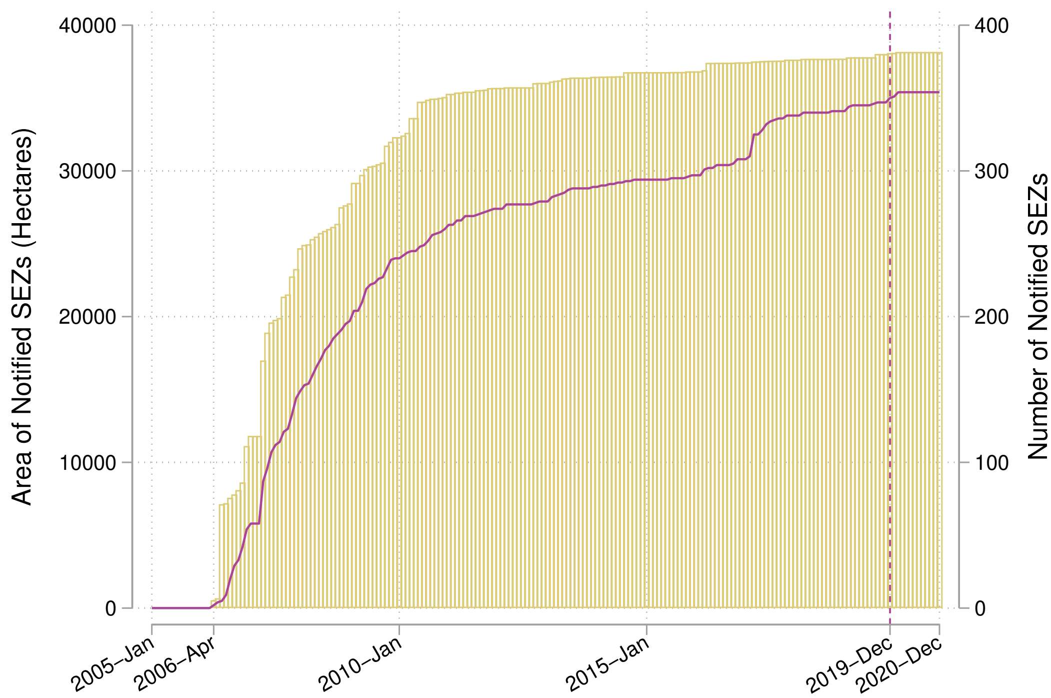
Table 6: Heterogeneity Results: By Region

	(1) Cluster 1	(2) Cluster 2
β	-0.256***	-0.169
	(0.092)	(0.271)
N	29143	4784
R^2	0.640	0.731

Notes: * p<0.10 ** p<0.05 *** p<0.01. Standard errors clustered at the district level are reported in parentheses. This table summarizes results for two regional clusters, factoring in India's energy distribution. The estimation is based on the preferred model (equation (3)) controlling for year, state, treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. Cluster 1 includes the Northern, Southern, and Western Zonals, where renewable energy is rapidly expanding and constitutes a relatively larger share of the total energy mix; Cluster 2 consists of North Eastern, Eastern, and Central Zonals, where renewable energy has a smaller presence.

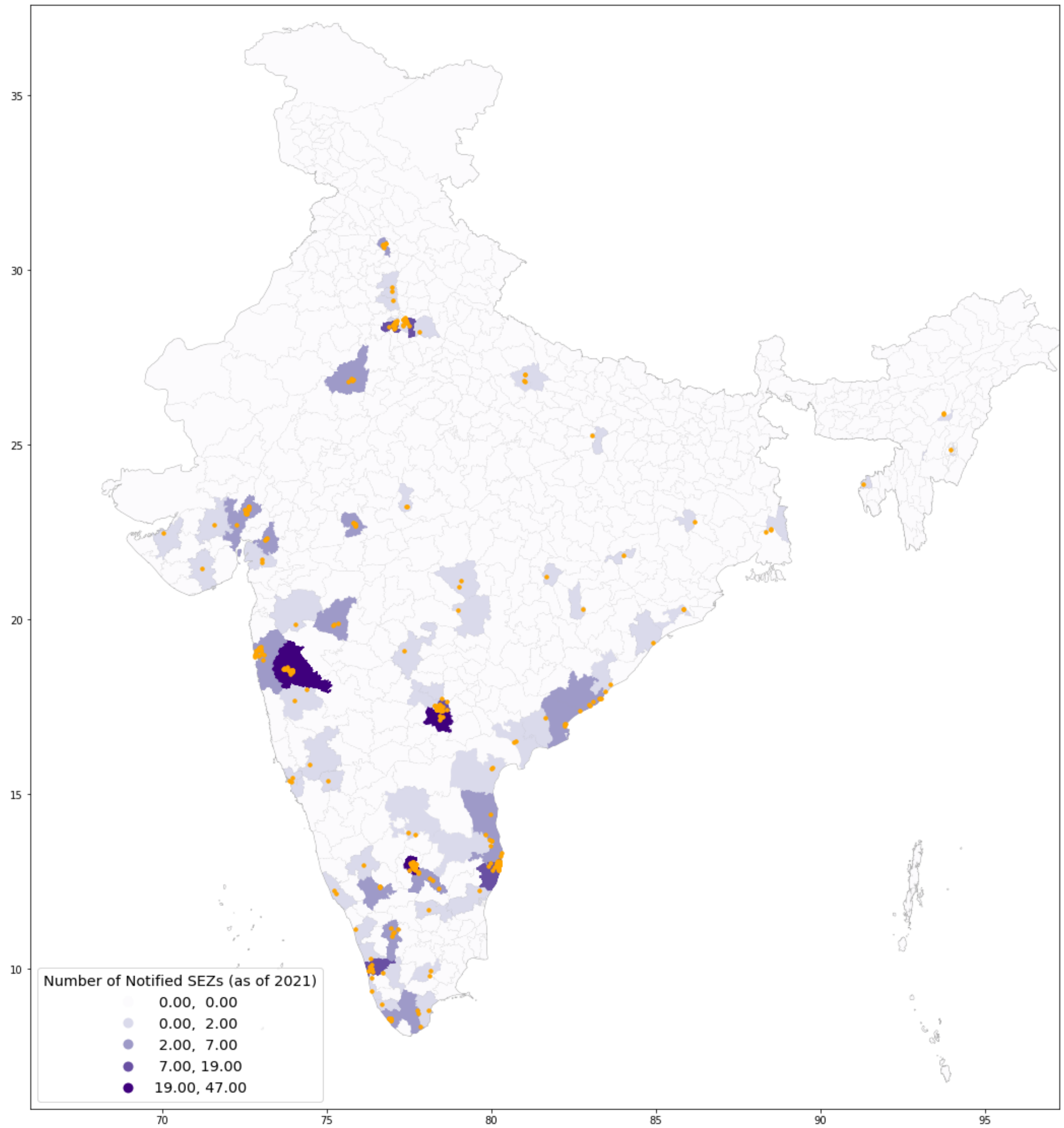
Online Appendix

A Appendix Figures



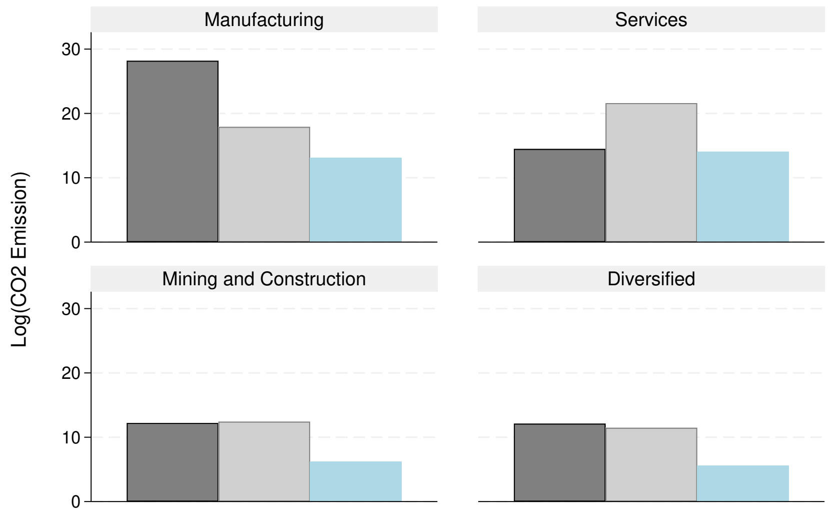
A.1: SEZs in India

Notes: This figure plots the evolution of the number of notified SEZs (purple line) and their areas (yellow bar) since the enactment of the SEZs Act in 2005.

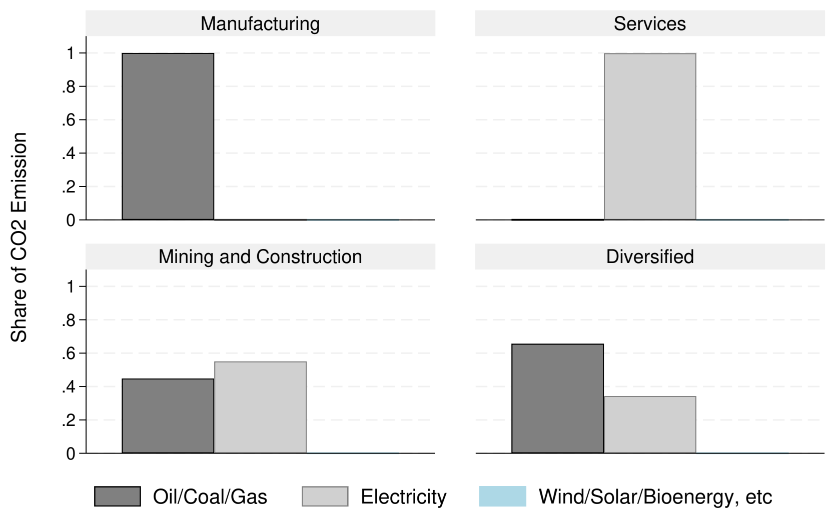


A.2: SEZs in India

Notes: This figure illustrates the geographic distribution of the notified SEZs across India. Districts with a greater number of notified SEZs are shaded darker. The orange markers denote location of individual SEZs across India.



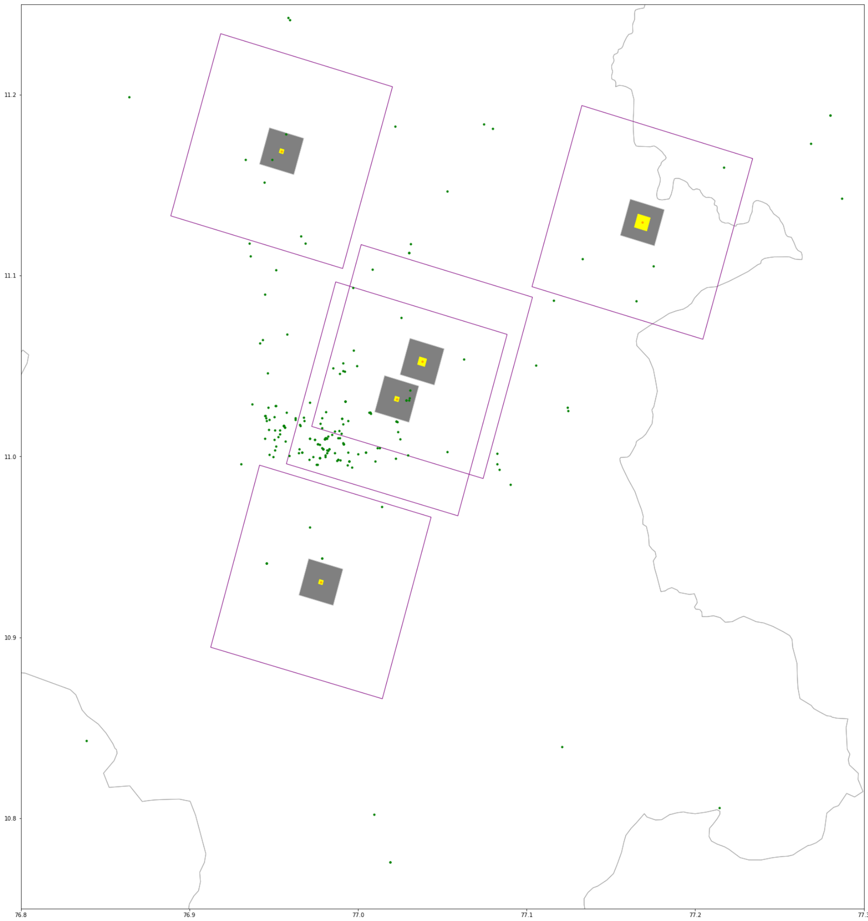
(a) Log Carbon Emission



(b) Share of Carbon Emission

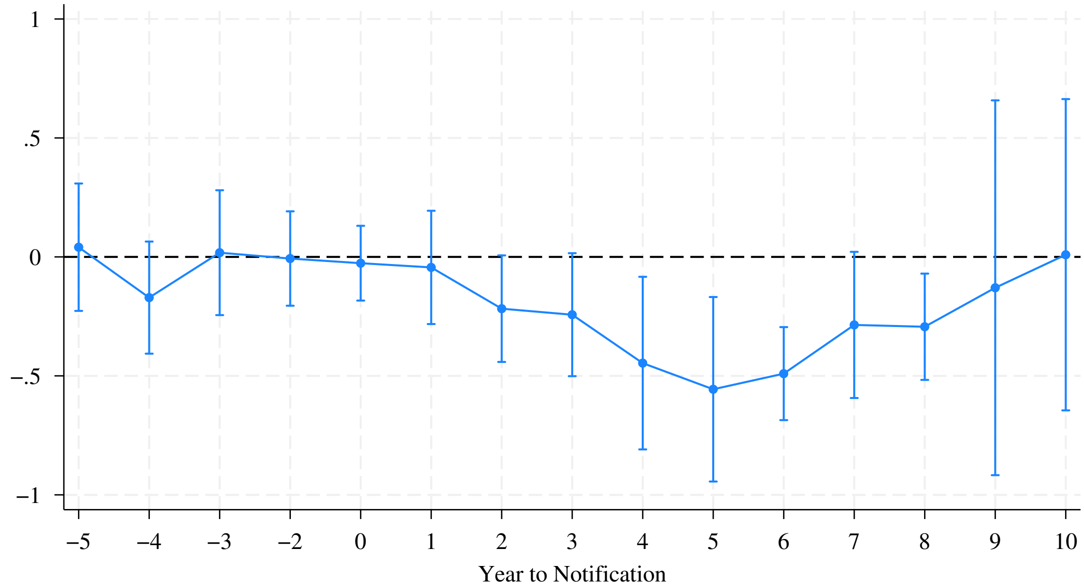
A.3: Carbon Emission by Industry and Fuel Type

Notes: This figure plots carbon emission by industry group and fuel type. Panel (a) reports log carbon emission, while Panel (b) reports the share of total carbon emission. We classify four industry groups as discussed in the main text: Manufacturing, Services (combining Financial and Non-Financial), Mining and Construction, and Diversified, and three broadly-defined energy categories: a) fossil fuels such as oil, coal, and natural gas, etc.; b) electricity; and c) renewable energy including wind, solar, and bioenergy, such as biogas or biomass, etc.



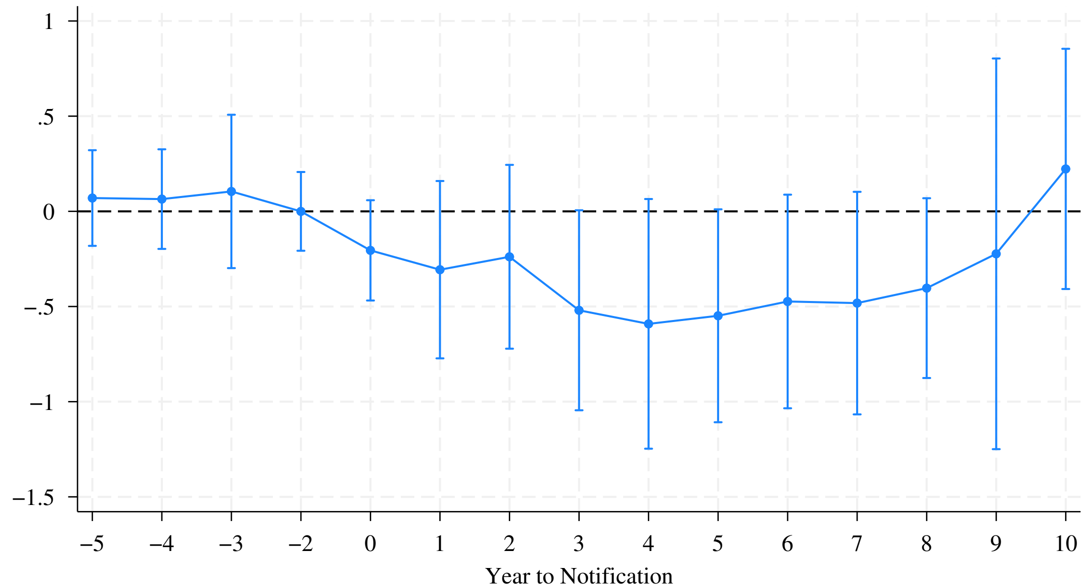
A.4: DiD Matching Example

Notes: This figure illustrates the treatment, control, and leave-out zones in the analysis. Yellow squares represent the treatment zones, which are the same size as reported in the SEZ notification. Grey areas denote the "leave-out" zones, equivalent to a 1 km $\times$ 1 km square. The analysis sample is restricted to firms within a 10 km $\times$ 10 km grid, indicated by the purple box.



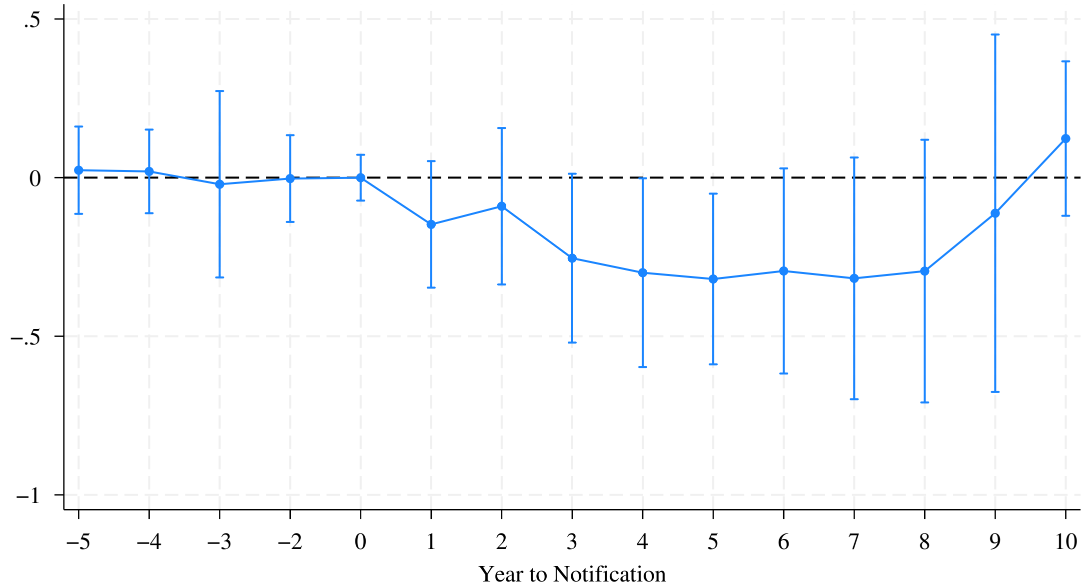
A.5: Event-Study Results: Energy Use

Notes: This figure presents the point estimates and their 95% confidence intervals for the event study estimating equation (4) where the dependent variable is log of annual energy consumption (in BTU) at the firm-year level. The model includes interaction terms between SEZ treatment and time dummies indicating 1-5 years before the SEZ notification and 1-10 years after the notification, and controls for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. The standard errors are clustered at the district level.



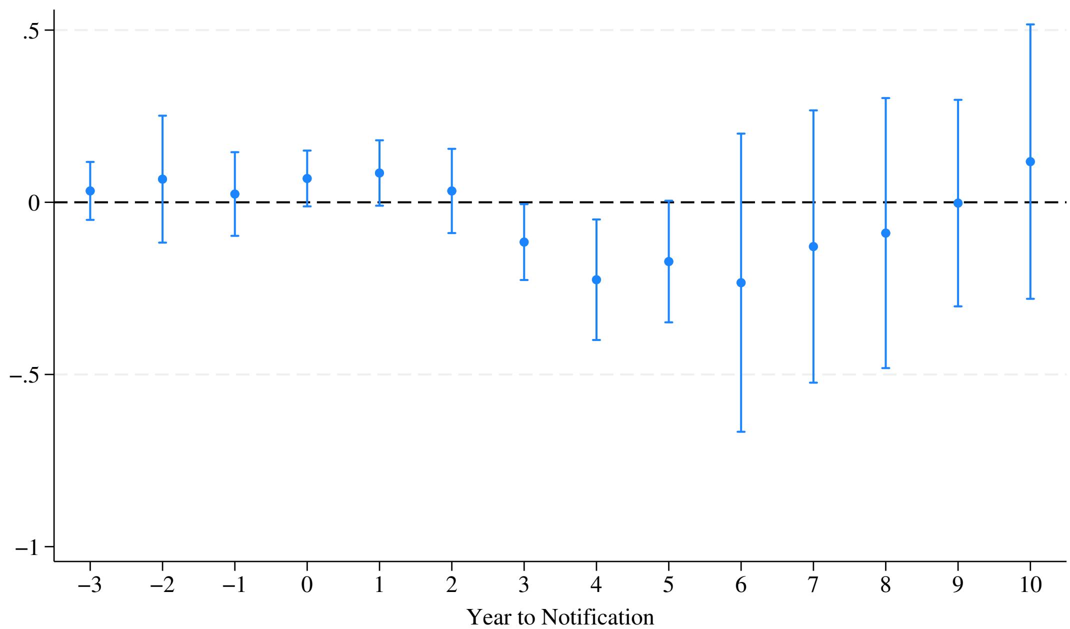
A.6: Event-Study Results: Energy Use per Sale

Notes: This figure presents the point estimates and their 95% confidence intervals for the event study estimating equation (4) where the dependent variable is log of annual energy consumption (in BTU) per sale at the firm-year level. The model includes interaction terms between SEZ treatment and time dummies indicating 1-5 years before the SEZ notification and 1-10 years after the notification, and controls for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. The standard errors are clustered at the district level.



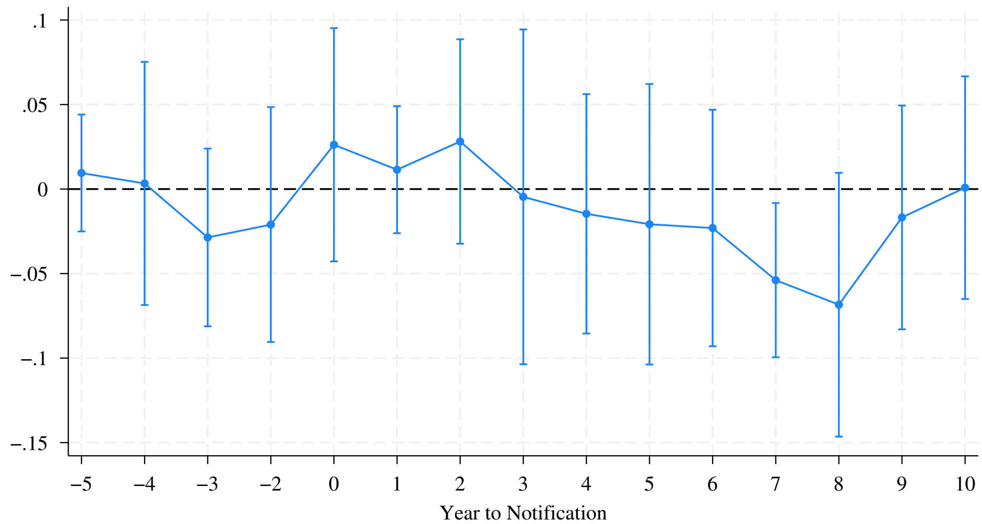
A.7: Event-Study Results: Emission per Sale

Notes: This figure presents the point estimates and their 95% confidence intervals for the event study estimating equation (4) where the dependent variable is log of annual carbon emissions per sale at the firm-year level. The model includes interaction terms between SEZ treatment and time dummies indicating 1-5 years before the SEZ notification and 1-10 years after the notification, and controls for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. The standard errors are clustered at the district level.



A.8: Robustness: Results of IPW

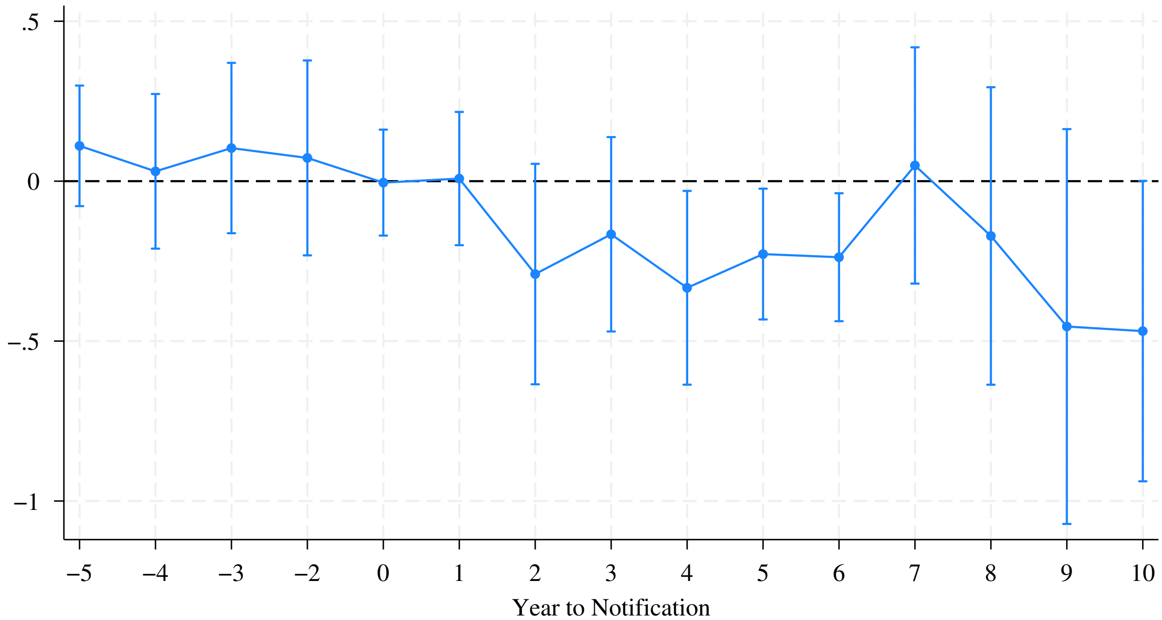
Notes: This figure plots the point estimates of equation (4) and 95% Confidence Intervals using the inverse probability weighting DiD estimator with stabilized weights. The y-axis measures the log of annual carbon emissions at the firm level. Results are estimated by the *csdid* command in Stata. Standard errors are clustered at the district level. Essentially it is equivalent to a two-way clustering at the panel and district level, see Callaway and Sant'Anna (2021) for more details. The control group is defined as these never treated.



(a) Event Study

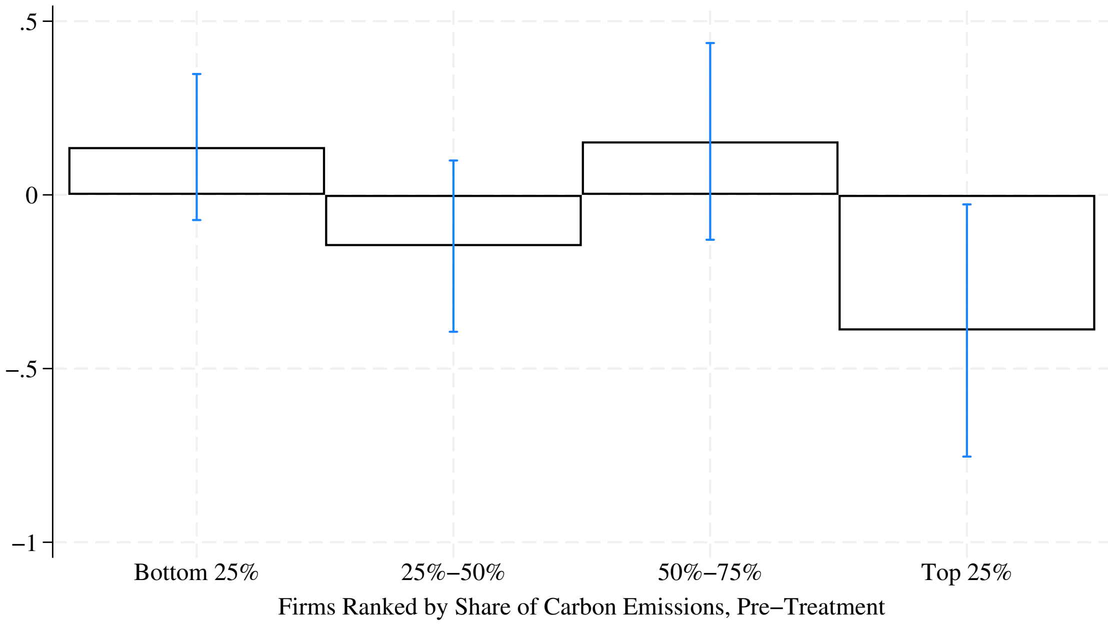
A.9: Examination of Reporting Accuracy

Notes: In this figure, we examine whether SEZ and non-SEZ firms exhibit different trends in accurately tracking their energy consumption before and after the establishment of SEZs. To capture potential inaccuracies in reporting, we treat missing values, arising either from ambiguous energy names and unit measures in the energy consumption table or from missing consumption quantities in the carbon emission calculations, as indicators of inaccurate reporting or insufficient tracking. Based on this, we construct a binary variable that equals one if any such missing values are present. We then replicate our baseline analysis using this binary indicator as the outcome variable.



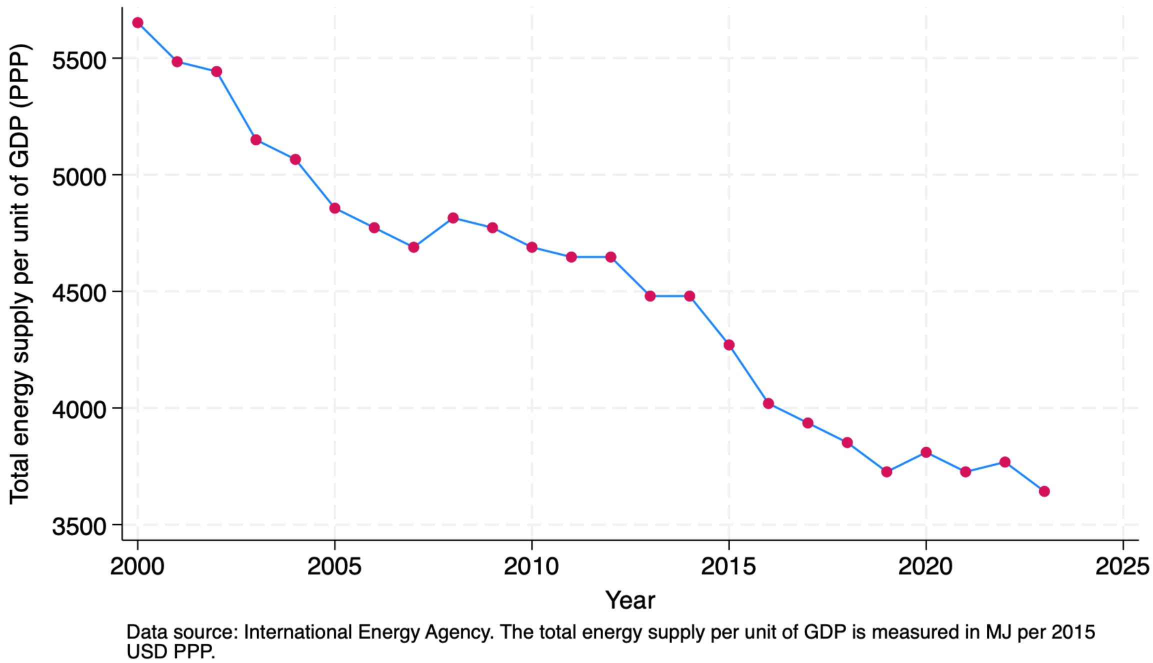
A.10: Additional Event-Study Results

Notes: This figure replicates the event study results from the main analysis using a sub-sample consisting only of firms that consistently and accurately tracked their energy use throughout the sample period. Point estimates and their 90% confidence intervals are reported. The new results remain qualitatively and quantitatively similar to those reported in the main text.



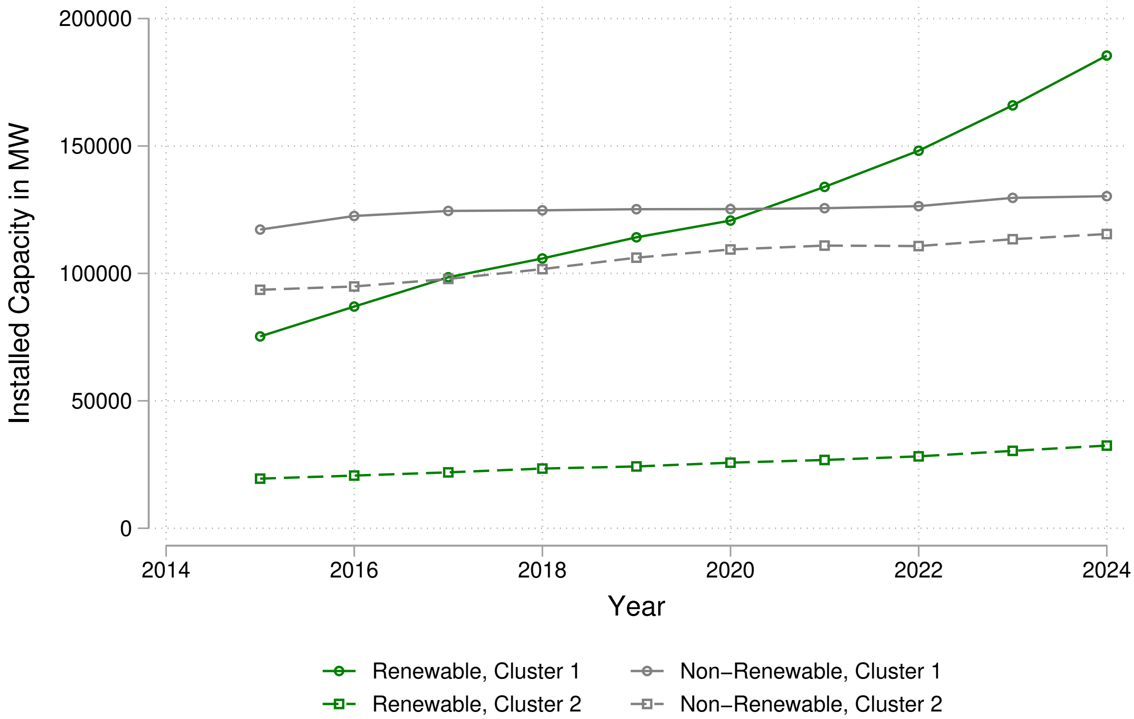
A.11: Marginal Effects of SEZs on Carbon Emissions by Emitter Type

Notes: This figure plots the heterogeneous effects of SEZs on the logarithm of carbon emissions by emitter type. Firms are ranked by their share of carbon emissions during the pre-treatment period and grouped into quartiles. Point estimates and 90% confidence intervals of the marginal effects are reported. The estimation results are based on the preferred model with standard error clustering at the district level.



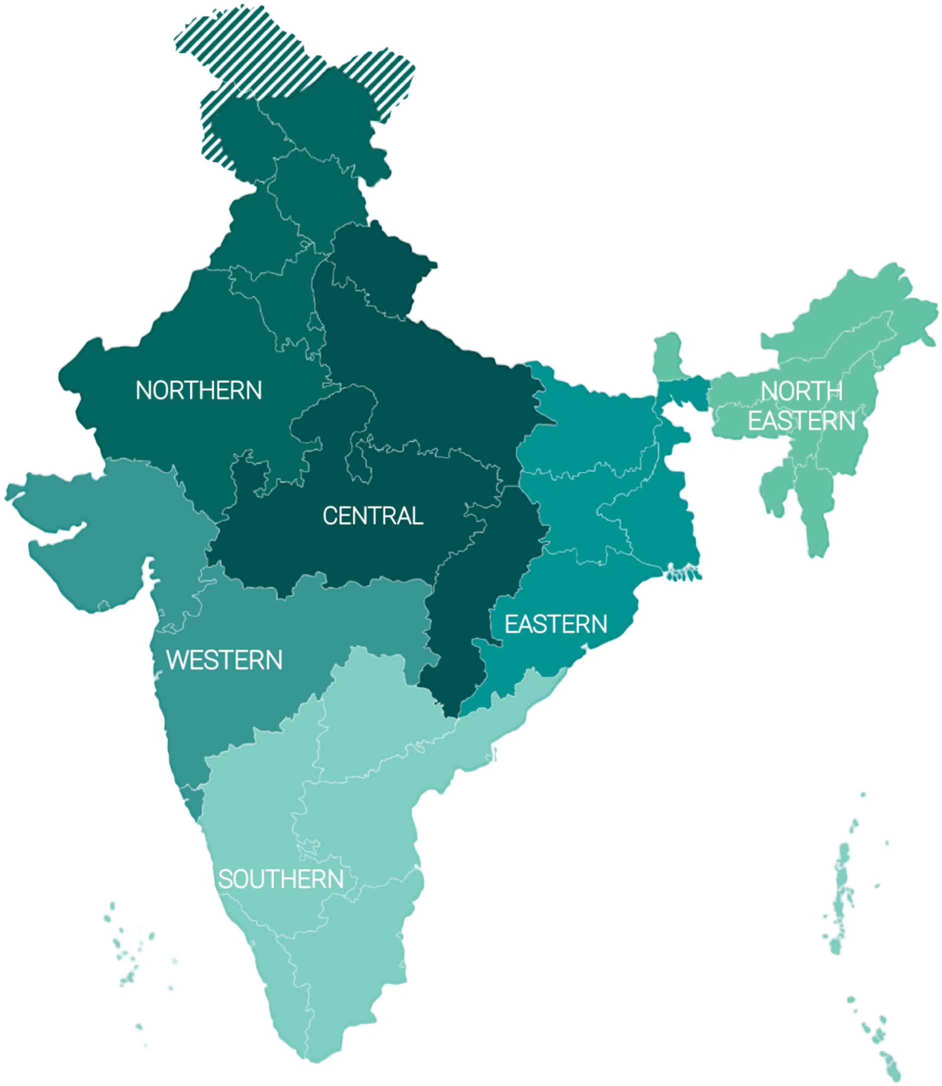
A.12: Energy Use per Unit of Output in India

Notes: This figure plots total energy supply per unit of GDP in India over time, measured in megajoules (MJ) per 2015 USD (PPP). Data are sourced from the International Energy Agency (IEA).



A.13: Installed Electricity Capacity (in MW) by Energy Source and Region

Notes: This figure shows the trend in installed electricity capacity (in MW) by energy source and region. Data are sourced from ICED: https://iced.niti.gov.in/energy. The regional cluster definitions are consistent with those used in the main analysis. The ICED data are only available from 2015 onward.



A.14: India Zonal Divisions

Notes: India zonal divisions figure source: https://en.wikipedia.org/wiki/Administrative_divisions_of_India. India's administrative regional (zonal) divisions split the country into six regions. They are Northern Zonal Council, comprising Chandigarh, Delhi, Haryana, Himachal Pradesh, Jammu and Kashmir, Ladakh, Punjab, and Rajasthan; North Eastern Council, comprising Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland and Tripura; Central Zonal Council, comprising the States of Chhattisgarh, Madhya Pradesh, Uttarakhand and Uttar Pradesh; Eastern Zonal Council, comprising Bihar, Jharkhand, Odisha, and West Bengal; Western Zonal Council, comprising Dadra and Nagar Haveli and Daman and Diu, Goa, Gujarat, and Maharashtra; Southern Zonal Council, comprising Andhra Pradesh, Karnataka, Kerala, Puducherry, Tamil Nadu, and Telangana.

B Appendix Tables

B.1: Carbon Emission Factors

Energy Source	kg CO2 per unit of Energy Source	Unit of Energy Source	kg CO2 per MMBtu of Energy Source
(1)	(2)	(3)	(4)
Agricultural Byproducts	975	short ton	118.17
Biodiesel (100%)	9.45	gallon	73.84
Biogas (Captured Methane)	0.044	scf	52.07
Bituminous Coal	2325	short ton	93.28
Coal Coke	2819	short ton	113.67
Coke Oven Gas	0.03	scf	46.85
Crude Oil	10.29	gallon	74.54
Distillate Fuel Oil No.1	10.18	gallon	73.25
Distillate Fuel Oil No.2	10.21	gallon	73.96
Fuel Gas	0.08	scf	59
Heavy Gas Oils	11.09	gallon	74.92
Kerosene	10.15	gallon	75.2
Lignite Coal	1389	short ton	97.72
Liquefied Petroleum Gases (LPG)	5.68	gallon	61.71
Lubricants	10.69	gallon	74.27
Mixed (Electric Power Sector)	1885	short ton	95.52
Mixed (Industrial Sector)	2116	short ton	94.67
Motor Gasoline	8.78	gallon	70.22
Naphtha (<401 deg F)	8.5	gallon	68.02
Natural Gas	0.05	scf	53.06
Natural Gasoline	7.36	gallon	66.88
Other Biomass Gases	0.03	scf	52.07
Peat	895	short ton	111.84
Petroleum Coke (Solid)	3072	short ton	102.41
Propane	5.72	gallon	62.87
Rendered Animal Fat	8.88	gallon	71.06
Residual Fuel Oil No.5	10.21	gallon	72.93
Residual Fuel Oil No.6	11.27	gallon	75.1
Solid Byproducts	1096	short ton	105.51
Wood and Wood Residuals	1640	short ton	93.8

Notes: This table summarizes the carbon emission factors by energy source used in this paper. The table is extracted from the EPA's GHG Emission Factors Hub (https://www.epa.gov/climateleadership/ghg-emission-factors-hub).

B.2: Summary Statistics

	(1)	(2)	(3)	(4)	(5)
Variable	Obs	Mean	Std. Dev.	Min	Max
CO2 Emission (Metric Ton)	80,070	21856.65	4722007	0	1,320,000,000
Log(CO2 Emission)	78,600	2.70	2.52	-14.03	27.91
CO2 Emission from Renewable	79,964	22.32	2277.12	0	443668.5
CO2 Emission from Conventional	79,909	21900	4730000	0	1,320,000,000
Share from Conventional	78,600	0.98	0.10	0	1
Log(CO2 Emission from Renewable)	80,070	0.13	0.74	-14.57	13.00
Log(CO2 Emission from Conventional)	80,070	2.62	2.52	-14.03	27.91
Log(Energy Consumption in MMBtu)	79,147	11.04	2.94	-4.51	37.52
Expenses	37,368	868.15	750.95	0.23	1565.00
Expenses (Fuel)	37,368	155.15	151.61	0.004	309.75
Sales	37,368	480.33	431.28	0	882.03
Profit	37,368	5.87	82.77	-2759.75	4497.27
Investment	37,368	71.75	13.85	0	74.46
Total Assets	37,368	1779.04	326.47	0.78	1842.50
PPE	37,368	701.63	128.42	0.07	726.51
Land and Building	37,368	185.89	34.41	0	192.58
Total Population, District (in 1 million)	79,765	4.36	2.64	0.14	11.06
Log(Total Population)	79,765	15.08	0.72	11.86	16.22
	Freq.	Percent			
Firm Age					
Before 1950	8,374	10.46			
Between 1951 and 1971	8,714	10.88			
Between 1972 and 1985	21,295	26.6			
Between 1986 and 1990	13,297	16.61			
After 1991	28,389	35.46			
Between 1991 and 2000	21,547	26.91			
Between 2001 and 2005	4,296	5.37			
After 2006	2,547	3.18			
Size by Decile					
Decile 1	13,391	16.72			
Decile 2	12,897	16.11			
Decile 3	11,233	14.03			
Decile 4	10,120	12.64			
Decile 5	9,724	12.14			
Decile 6	8,128	10.15			
Decile 7	5,808	7.25			
Decile 8	4,186	5.23			
Decile 9	2,953	3.69			
Decile 10	1,606	2.01			

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Continued

	Freq.	Percent
Entity type		
Associations/Federations	9	0.01
Co-operatives	7	0.01
Departmental undertakings/Boards	2	0
Foreign Entities	1	0
Governments	18	0.02
Partnership firms	8	0.01
Private Ltd.	14,559	18.18
Public Ltd.	65,450	81.74
Trusts	7	0.01
Unlimited Liabilities	9	0.01
CMIE 6-digits		
Manufacturing	64,800	80.93
Mining	697	0.87
Electricity	197	0.25
Services (other than financial)	8,514	10.63
Construction & real estate	744	0.93
Asset financing services	40	0.05
Other fund based financial	3,663	4.57
Fee based financial services	2	0
Other financial services	24	0.03
Diversified financial services	234	0.29
Diversified	1,155	1.44
N	80070	

Notes: This table reports the summary statistics for the analysis sample. Energy consumption is measured in million British Thermal Unit (BTU). Expenses, profit, and sales, investment (short-term), total assets, PPE (property, plant, and equipment), and land and building are measured in million US dollars.

B.3: Summary Statistics by Treatment Status

	(1)	(2)	(3)	(4)	(5)	(6)
	Treatment Group		Control Group		Control-Treatment	
	Mean	Std.	Mean	Std.	Diff.	t-stats
Log(CO2 Emission)	2.483	2.695	2.695	2.501	0.212***	(5.947)
Firm Age	3.555	1.248	3.560	1.361	0.005	(0.277)
Size Decile	3.996	2.491	4.207	2.430	0.211***	(6.437)
Entity Type	2.136	0.349	2.189	0.392	0.053***	(11.329)
Industry Group						
Manufacturing	0.781	0.414	0.815	0.388	0.034***	(6.306)
Non-Financial Services	0.131	0.338	0.106	0.308	-0.025***	(-5.669)
Mining and Construction	0.020	0.141	0.017	0.128	-0.004*	(-1.989)
Financial Services	0.054	0.226	0.046	0.210	-0.008**	(-2.613)
Diversified	0.014	0.117	0.016	0.126	0.002	(1.430)
N	6328		62426		68754	

Notes: $*p < 0.10 **p < 0.05 ***p < 0.01$. This table reports the summary statistics by treatment status and the results of t-test for the difference between the two groups.

B.4: Differences in Firm Attributes by Treatment Status, Matched v.s. Unmatched Sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Firm Age	Size Decile	Entity Type	Industry Group Indicator				
				Manufacturing	Non-Financial Services	Mining and Construction	Financial Services	Diversified
	Unmatched Sample							
SEZs	-0.0046	-0.211***	-0.053***	-0.0342***	0.0251***	0.0037**	0.0077***	-0.0022
	(0.0166)	(0.0328)	(0.0047)	(0.0054)	(0.0044)	(0.0019)	(0.0030)	(0.0016)
N	68753	68730	68754	68754	68754	68754	68754	68754
Pair FEs	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
	Matched Sample							
SEZs	-0.0063	0.0042	0.0244	-0.0354	0.0354	0.0004	-0.0018	0.0014
	(0.0412)	(0.0585)	(0.0301)	((0.0349)	(0.0275)	(0.0165)	(0.0167)	(0.0147)
N	38817	38817	38817	38817	38817	38817	38817	38817
Pair FEs	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*

Notes: * p<0.10 ** p<0.05 *** p<0.01. This table reports the results regressing firm attributes on the within SEZs indicator separately for matched and unmatched samples, where for the matched sample, pair FEs are included. Standard errors are reported in parentheses. For the unmatched sample, we report robust standard errors, as the model includes no fixed effects or control variables, and the estimates simply reflect the sample mean difference between the treatment and control groups. For the matched sample, we report clustered standard errors at the treatment-control pair level.

B.5: Observable Characteristics of New and Incumbent Firms Within SEZs

	New Firms		Incumbent Firms		Incumbent - New	
	Mean	Std.	Mean	Std.	Diff.	t-stats
Firm Age Cohort	5.000	0.000	3.795	1.231	-1.205***	(-48.295)
Size Decile	3.310	2.202	3.666	2.206	0.355	(1.214)
Industry Group						
Manufacturing	0.862	0.348	0.811	0.392	-0.051	(-1.101)
Non-Financial Services	0.086	0.283	0.129	0.336	0.043	(1.143)
Mining and Construction	0.000	0.000	0.011	0.105	0.011***	(5.224)
Financial Services	0.052	0.223	0.039	0.194	-0.013	(-0.429)
Diversified	0.000	0.000	0.009	0.097	0.009***	(4.818)
Entity Group						
Public	0.603	0.493	0.787	0.410	0.183**	(2.806)
Private	0.397	0.493	0.209	0.406	-0.188**	(-2.876)
Others	0.000	0.000	0.005	0.067	0.005***	(3.323)
N	58		2434		2492	

Notes: This table compares observable characteristics of new firms and incumbent firms located within SEZs. New firms are defined as firms established after the SEZ notification, while incumbent firms are defined as firms established before the SEZ notification. Both groups are restricted to firms located within SEZs. The establishment cohort index is coded from 1 to 5 based on firms' establishment periods, where 1 denotes firms established before 1950 and 5 denotes firms established after 1991. A larger value of the establishment cohort index therefore indicates a more recently established, or younger, firm. The difference column reports the mean difference between incumbent and new firms, calculated as incumbent minus new. The corresponding t-statistics are reported in parentheses. $*p < 0.10 **p < 0.05 ***p < 0.01$.

B.6: Additional Results: Total Carbon Emissions and Energy Consumption (BTU)

	(1)	(2)	(3)	(4)
	Log(Energy Use/Sale)	Log(Carbon Emissions/Sale)	Log(Energy Use)	Log(Carbon Emissions)
β	-0.353***	-0.127*	-0.418***	-0.393***
	(0.1112)	(0.0697)	(0.1482)	(0.1352)
N	16125	16125	16056	15965
R^2	0.57	0.50	0.73	0.74

Notes: This table shows the main estimation results using the spatial RD-DiD design. * p<0.10 ** p<0.05 *** p<0.01. Standard errors clustered at the district level are reported in parentheses. All models control for treatment-control pair, industry (8-digit CMIE industry codes), state by year, and district fixed effects. Energy consumption is measured in millions of British Thermal Units (BTU).

B.7: Robustness Check

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Log(Carbon Emissions)							
β	-0.248**	-0.248***	-0.248**	-0.247***	-0.249***	-0.203**	-0.328**	-0.288**
	(0.110)	(0.059)	(0.105)	(0.076)	(0.090)	(0.092)	(0.126)	(0.133)
N	33932	33932	33932	30729	32745	28239	22177	16482
R^2	0.640	0.640	0.640	0.662	0.648	0.657	0.696	0.717
Cluster-Level	Pair	State + Year	Firm + District-Year					
Sub-Sample				Inc. Yr $\leq$ 2000	Inc. Yr $\leq$ 2005			
Sample Zone	10 km	10 km	10 km	10 km	10 km	10 km	5 km	5 km
Leave-out Zone	1 km	1 km	1 km	1 km	1 km	2 km	1 km	2 km

Note: This table summarizes the results of robustness check. Standard errors in parentheses. Unless otherwise specified, standard errors clustered at the district level are reported. * $p < 0.10$, **$p < 0.05$, *** $p < 0.01$. Columns (1) to (3) report the results with standard errors clustered at treatment-control pair, state and year, and district and year levels, respectively. Columns (4) and (5) re-estimate the preferred model on sub-samples of firms incorporated before 2000 or 2005. Columns (6) to (8) experiment with different sample zones, such as a 5 km zone, and different leave-out zones, such as a 1 km zone.

C Model Derivation

Solve for the optimal total energy consumption Here, we formally derive the optimal energy consumption, by solving the firm's optimization problem. First, firms choose the optimal production level Y to maximize profits:

$$\max_{Y_s} \Pi = (1 - t(1-c))pY_s - C(Y_s, P)$$

and the optimal input levels to minimize costs:

$$\min_{K,E,L} C(Y_s, P) = p^K \cdot K_s + p^E \cdot E_s + p^L \cdot L_s,\ s.t.\ Y_s \geq \bar{Y}_s$$

Solving the first-order conditions of the cost minimization problem, we obtain the optimized cost function:

$$C^*(Y_s, P) = \left(\frac{Y_s}{A'_s}\right)^{\frac{1}{\psi}} \left(\frac{p^K}{\psi^K}\right)^{\frac{\psi^K}{\psi}} \left(\frac{p^L}{\psi^L}\right)^{\frac{\psi^L}{\psi}} \left(\frac{p^E}{\psi^E}\right)^{\frac{\psi^E}{\psi}} \psi$$

and the conditional energy demand:

$$E^*(\bar{Y}_s, P) = \left(\frac{\bar{Y}_s}{\tilde{A}_s}\right)^{\frac{1}{\psi}} \left(\frac{p^K}{\psi^K}\right)^{\frac{\psi^K}{\psi}} \left(\frac{p^L}{\psi^L}\right)^{\frac{\psi^L}{\psi}} \left(\frac{p^E}{\psi^E}\right)^{\frac{\psi^E - \psi}{\psi}}$$

where $\psi = \psi^K + \psi^E + \psi^L$ and $\tilde{A}_s = A_s (A_s^E)^{\psi_E}$. We can see from the conditional energy demand function that holding the production level constant, a higher energy efficiency (i.e., a larger A_s^E) reduces energy demand. Plugging this into the profit function and solving the first-order condition gives the optimal output level:

$$Y_s^* = \tilde{A}_s^{\frac{1}{1-\psi}} \Big[(1 - t(1-c))p\Big]^{\frac{\psi}{1-\psi}} \cdot \left(\frac{\psi^K}{p^K}\right)^{\frac{\psi^K}{1-\psi}} \cdot \left(\frac{\psi^L}{p^L}\right)^{\frac{\psi^L}{1-\psi}} \cdot \left(\frac{\psi^E}{p^E}\right)^{\frac{\psi^E}{1-\psi}}$$

The corresponding optimal energy demand is:

$$E_s^* = (Y_s^*)^{\frac{1}{\psi}} \tilde{A}_s^{-\frac{1}{\psi}} \left(\frac{p^K}{\psi^K}\right)^{\frac{\psi^K}{\psi}} \left(\frac{p^L}{\psi^L}\right)^{\frac{\psi^L}{\psi}} \left(\frac{p^E}{\psi^E}\right)^{\frac{\psi^E - \psi}{\psi}}$$

□

Solve for the optimal energy consumption across types After determining the optimal level of total energy consumption, firms in sector s decide how to allocate this energy between cleaner and conventional sources. Specifically, they choose the quantities of energy inputs E_s^x, where $x \in \{c, d\}$

denotes clean and dirty energy, respectively, in order to minimize total energy costs:

$$\begin{aligned} \min_{E_s^c, E_s^d} \quad & p_s^{E^d} E_s^d + p_s^{E^c} E_s^c + f\, \mathbb{1}(E_s^c > 0) \\ \text{s.t.} \quad & \begin{cases} E_s^* = \delta_c E_s^c + \delta_d E_s^d, & s = N \\ E_s^* = \left[\delta_c (E_s^c)^\rho + \delta_d (E_s^d)^\rho\right]^{\frac{1}{\rho}}, & s = M \end{cases} \end{aligned} \tag{5}$$

First, consider non-manufacturing firms, where clean and dirty energy sources are perfectly substitutable. In this case, corner solutions arise: a firm will adopt only cleaner energy when the following condition holds:

$$(Y_N^*)^{\frac{1}{\psi}} > f \cdot \left(\frac{\delta_c \delta_d}{p^{E^d}\delta_c - p^{E^c}\delta_d}\right) \cdot \tilde{A}_s^{\frac{1}{\psi}} \cdot \Psi \tag{6}$$

where $\Psi = \left(\frac{p^K}{\psi^K}\right)^{-\frac{\psi^K}{\psi}} \left(\frac{p^L}{\psi^L}\right)^{-\frac{\psi^L}{\psi}} \left(\frac{p^E}{\psi^E}\right)^{-\frac{\psi^E - \psi}{\psi}}$, which collects the price and elasticity parameters appearing in the optimal output function. We show below that this is not the case for manufacturing firms. Solving the first-order condition of the optimization problem in equation (5), we obtain:

$$\frac{E_M^{c*}}{E_M^{d*}} = \left(\frac{(1-\delta_c)p^{E^c}}{\delta_d p^{E^d}}\right)^{\frac{1}{\rho_M - 1}} \quad \text{and} \quad \frac{E_M^{c*}}{E_M^*} = \frac{\gamma}{(\delta_c \gamma^{\rho_M} + \delta_d)^{1/\rho_M}}$$

where $\gamma = \left(\frac{(1-\delta_c)p^{E^c}}{\delta_d p^{E^d}}\right)^{\frac{1}{\rho_M - 1}}$. □

D Calculation of Firms' Carbon Emissions from Energy Consumption

In this appendix, we provide additional details on the data sources and computation procedures used to construct firm-level carbon emission data based on firms' energy consumption.

Firm-level energy use data comes from *Prowess*. *Prowess* contains information on the annual total energy consumption reported by energy source, each with its corresponding unit. For example, a firm that consumed electricity and coal in a given year would have two separate records: one for the number of kilowatt-hours (kWhs) of electricity and another for the number of metric tonnes of coal.

To calculate the annual total carbon emissions from the total energy consumption, similar to Barrows and Ollivier (2018), we assign a source-specific carbon dioxide emission factor to each energy source type.[19] Several steps involved in this process. First, we match and create a crosswalk between the energy source types reported in *Prowess* and the energy categories listed in the EPA's emission factors table. 127 over 148 (86%) energy sources in the *Prowess* are matched to the emission factors table. Then, we standardize the unit of measurement for each type of energy source with the unit of the corresponding emission factor and assign a conversion factor for each energy source-unit pair. We are unable to standardize the energy units for 182 energy source-unit pairs (over 852 total pairs) due to missing observations of the measurement unit and typos in the measurement unit. For example, in some cases, the unit of coal is measured in "Liters" or "Kiloliters", which are volume measures typically used for liquids or fluids. Without knowing the density of the coal being used, it is difficult to convert a volume measure to a weight measure.

It is worth noting that instead of assigning a single carbon emission factor to the energy source type "electricity" (e.g., Barrows and Ollivier 2018), we distinguish electricity purchased by the firm and electricity generated by the firm from various sources, such as coal, gas and oil, and biomass fuels. Emissions from electricity generation vary by types of energy source. For example, electricity generated by the so-called green energy, such as solar and wind, produces zero direct carbon emission, while electricity generated by conventional fuels, such as coal and natural gas, emits carbon dioxide ranging from 0.35 kg/kWh to 0.87 kg/kWh (Schlömer et al. 2014), depending on the type and efficiency of electric power plants. In other words, electricity generated by solar or wind and electricity generated by coal or natural gas are essentially two energy products in terms of carbon emissions and should be recognized as two different energy source types.

To capture this feature, we adopt an additional procedure when calculating emissions from electricity consumption, accounting for variations in the energy composition (i.e., fossil fuel-based electricity vs. non-fossil-based electricity) used in electricity generation across states and years. Specifically, in our approach, we distinguish between purchased and self-generated electricity. For self-generated electricity, the consumption records in the data specify a single generation source. The generation sources include both conventional fuels, such as coal, oil, and gas, and renewable fuels, such as biomass, solar, and wind. We assign different carbon factors to electricity generated from each source. To obtain these carbon emission factors, we first collect data on India's electricity generation by energy source and fuel consumption from the India Climate and Energy Dashboard (ICED).[20] These data allow us to calculate the average amount of fuel needed to generate one kWh of electricity in India. Then, we transform the amount of fuel to the heating content (measured

[19]Following existing literature, we adopt the emission factors provided by the US Environmental Protection Agency (EPA). Appendix Table B1. summarized these factors.

[20]Data source: https://iced.niti.gov.in.

in million British thermal units, MMBtu) and multiply it by the carbon emission factor of the corresponding fuel (which is measured in kg/MMBtu). For example, in India, the average amount of coal needed to generate one kWh of electricity is about 1.54 pound (averaged over years from 2012 to 2019), with a heating value equivalent to about 0.0219 MMBtu. The carbon emission factor for coal used in electric power sector is 95.52 kg/MMBtu. By multiplying the number of coals needed by 95.52, we can obtain the carbon emissions for one kWh electricity generated by coal. We conduct this calculation separately for each fuel by year, to capture the temporal variation in the efficiency of electricity generation in India.[21] Note that in this calculation, we assume that the electricity generation within a firm is as efficient as in a representative electric power plant with average efficiency. It is likely that the efficiency of electricity generation is lower than that of a power plant, possibly due to differences in technology and a lack of economic scale. In that case, our calculation would provide a lower bound estimation of the carbon emissions for electricity generated by firms from various energy sources. For electricity generated from solar or wind, the carbon emission factor is set to zero, according to IPCC's estimate.

For purchased electricity, it is unclear whether the consumed electricity is generated from fossil or non-fossil fuels, as these sources are aggregated within the power grid. Therefore, the corresponding carbon emissions per unit of electricity consumption may vary across states, reflecting differences in the energy composition of their power systems. Considering this, we collect state-year level data on electricity generation by power source (including both fossil fuel and non-fossil fuel categories) from ICED. The ICED dataset provides information on the annual amount of electricity generated across states and the corresponding amounts of fuel consumed. This information allows us to compute both the share of electricity generated from each source and the average amount of fuel from each source used to produce one unit of electricity across states. Using these metrics, we then calculate the carbon emissions associated with purchased electricity as a weighted average of the carbon emissions from generating one kWh of electricity by energy source, with the shares of electricity generation by power source serving as the weights. Notably, variation in the carbon emissions of purchased electricity arises from two sources: (1) temporal and spatial variation in the shares of energy sources used for electricity generation, and (2) variation in the average generation efficiency for producing one unit of electricity.

In total, we have 216,686 observations of the firm-energy source-year pair for 8256 unique firms spanning 1988 to 2021. There are 146 observations missing the quantity of energy consumption and 13368 observations for which we cannot assign a carbon emission factor. We drop those observations, which consists of about 6% of the data.

[21] Since the electricity generation data is only available from 2012 to 2019, we approximate the amount of fuel needed for generating a kWh of electricity before 2012 using the 2012 statistics.